

CIF Operation Gridded Dataset of Daily Precipitation over Latin America (LatAm Dataset)

IRI Deliverable for Activity 2: Assessment of Diagnostic Results from Validation Study

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April 27th, 2018

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1 Preliminary Text

This report has been developed by the Financial Instruments Team at the International Research Institute for Climate and Society at Columbia University in New York, with support from the Insuresilience Investment Fund Technical Assistance Facility and in collaboration with the Instituto Nacional de Pesquisas Espaciais. If you have any questions please contact: Sofía Martínez (sms@iri.columbia.edu), Manuel Brahm (mbc@iri.columbia.edu) and / or Daniel Osgood (deo@iri.columbia.edu)

2 Overview

In this document the IRI presents the initial validation results conducted based on benchmarked dataset diagnostics. Initial findings of the validation of the *Operation Gridded Dataset of Daily Precipitation over Latin America (CoSch)*, developed by colleagues at the Instituto Nacional de Pesquisas Espaciais, hereon referred to as *LatAm Dataset* are being shared in this document. Overall, the validation study shows how this dataset can be useful in designing and validating index insurance products across the region, based on the needs of end-users.

Using tools and information that IRI has developed and gathered through IIF related and other efforts, we explored how an index developed with this new dataset would compare to indexes based on other types of remotely sensed data across Central and South America.

IRI used different data sources to compare the LatAm Dataset to the farmers´ perception in the field and other climate risk/events data, assessing its performance for index insurance purposes.

2.1 Implications in index design

Throughout an index insurance design process there are important trade-offs when selecting a dataset on which payouts are triggered. For example, some of the most critical are dataset latency, spatial scale and accuracy capturing losses on the ground. These trade-offs are context-specific and must be analyzed taking into consideration the end-users and implementer's needs. To develop index insurance products that have satisfactory performance, the IRI has developed and implemented processes that allow for comparisons across datasets and against losses reported on the ground.

In this report, the IRI focused on validating the performance of the LatAm dataset capturing insurance-relevant losses on the ground across Central and South America. Using a diverse set of statistical tools, we show that the LatAm dataset is a comparable source to other datasets currently being used for operational index insurance products.

Specifically in the countries where the IRI has gathered the most robust on-the-ground data (through several field research iterations,) such as Honduras, Uruguay and Colombia, we expect to see generally better performing than in other countries (Chile, Bolivia) where the project haven't reach that maturity stage.

Considering the other two key trade-offs, the choice of an operational index dataset will depend on the spatial scale and latency based on the specific type of crop and risk to be covered. For example, an index insurance product for dairy production might require a dataset with a short latency given the continuous nature of the productive system, however understanding that drought affects pasture in widespread areas the spatial scale might not need to be as refined. Conversely, if the weather patterns of a specific region have high variations, due to for example topography, then the spatial scale will be key in defining the final dataset option.

Overall looking at LatAm in comparison with the other satellite data sources, we see it is competitive, performing slightly better or slightly worse depending on the regions. Generally the results show that CHIRPS has a strong performance, with the LatAm dataset being competitive, with better results in some applications and worse in others. We can see that LatAm is outperformed slightly by CHIRPS in the excess rainfall application in all three countries, as well as in the drought application for Bolivia and Dominican Republic. LatAm is outperforming the other dataset in Colombia and Uruguay for drought and has similar results than CHIRPS in Chile and Honduras. TRMM is not as strong as the others in the majority of places, only seems strongest for Colombia's excess application.

The results from this validation study show that the LatAm dataset is competitive with other data sources and has strong performance at capturing the insurance-relevant losses on the ground, moreover an index design process should look further into the other trade-offs to achieve a design that is well-adjusted to the local needs. It is important to note that the analysis carried out here are coarse metrics, and that in the innovation processes in designing indexes, increased performance may be uncovered for the dataset (or other datasets) in each specific case.

3 Validation Methods

We focus on the principal of corroborating evidence in validation of the data source. Through diagnostic and more formal statistical methods we test how well the dataset predicts evidence of bad years across case studies. This evidence comes from farmer/expert interviews, other sources of climate risk events and alternate remote sensing sources. If there is a strong level of agreement between the data set and the corroborating evidence, we have confidence that the data source is valuable for index insurance design processes. When the agreement is lower, we investigate the mechanisms through which legitimate disagreements may exist, to be factored into the evaluation of the data source.

We performed three families of analysis:

1. Binary matching,
2. Logit regression to test the predictive skill of the dataset for farmer/expert bad years, and
3. Statistical comparison against other satellite data sources.

These analysis were carried out at two different spatial scales to evaluate how increasing the size of the area could change the performance of the datasets. The first spatial resolution was approximately 625 km^2, corresponding to the pixel size of the LatAm dataset, while the second spatial resolution is 5,625 km^2, corresponds to 3 by 3 pixels. Overall similar results were obtained in the two different spatial resolutions.

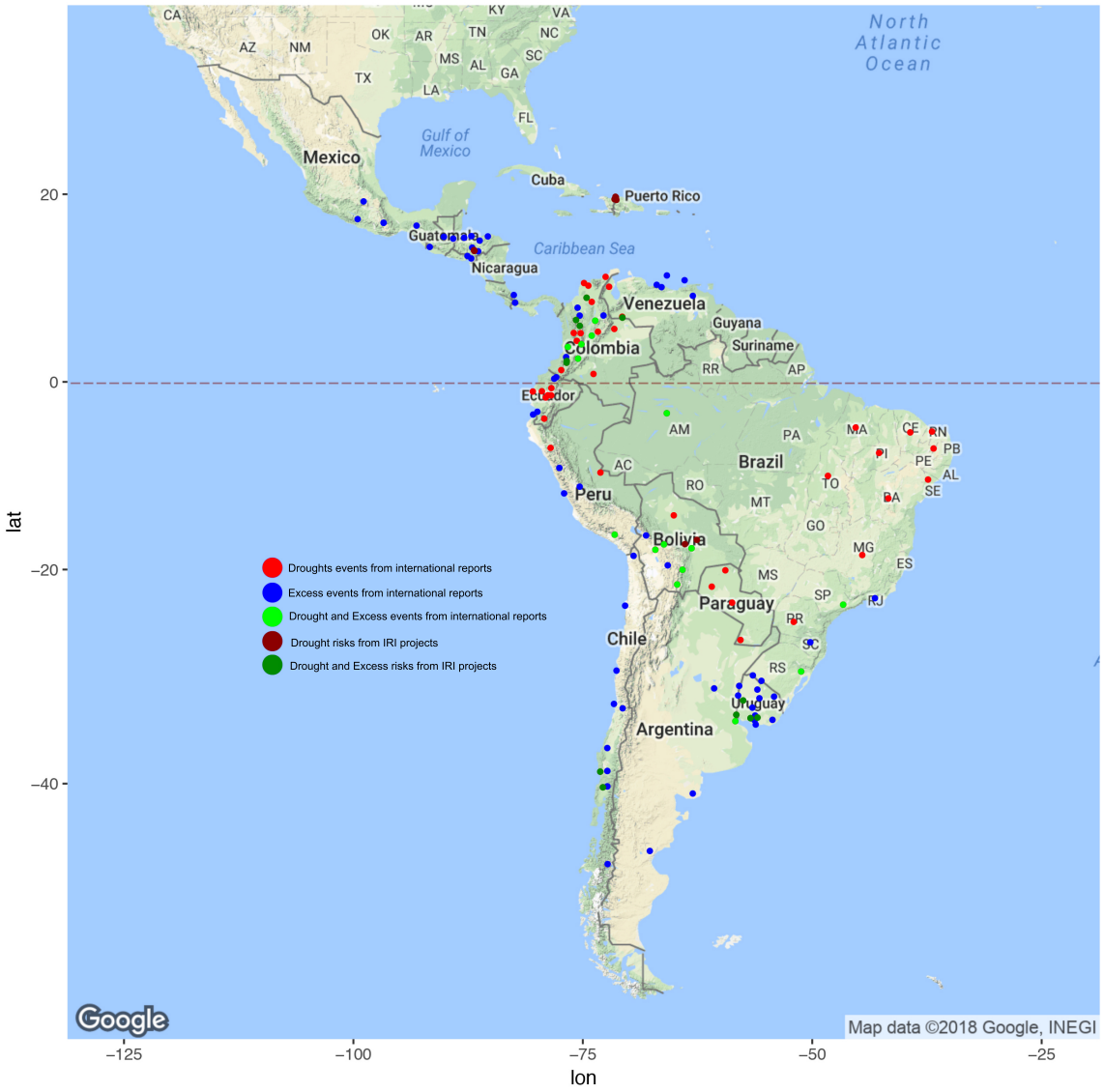
3.1 Binary Matching

Two different analysis were performed using the binary matching approach depending on the data source: Farmer bad years and other sources.

Because outliers generally determine the bulk of statistical correlations, and outliers are particularly important in the context of insurance, which is a tool specifically intended to target outlier years[1], an extremely valuable metric is the ability of a data source to target "bad years". By the nature of this data exercise, we are likely to be missing some important bad years from the list. Therefore our focus is more on testing if the data source flags known bad years than if it targets years that are not know as bad, since we do not know if the error is in accuracy, or the accidental omission of a bad year from the dataset. Nevertheless, to assess the performance of each of the datasets, the Heidke Skill Score (HSS) was calculated. The HSS measures the fractional improvement of the forecast over the standard forecast. Like most skill scores, it is normalized by the total range of possible improvement over the standard, which means Heidke Skill scores can safely be compared on different datasets. The range of the HSS is $-\infty$ to 1. Negative values indicate that the chance forecast is better, 0 means no skill, and a perfect forecast obtains a HSS of 1[2].

[1] Using Satellites to Make Index Insurance Scalable: Final IRI Report to the ILO Microinsurance Innovation Facility. September 8, 2014. IRI Columbia University.
[2] http://www.eumetrain.org/data/4/451/english/msg/ver_categ_forec/uos3/uos3_ko1.htm

Map 1: Study locations by type of risk and type of data



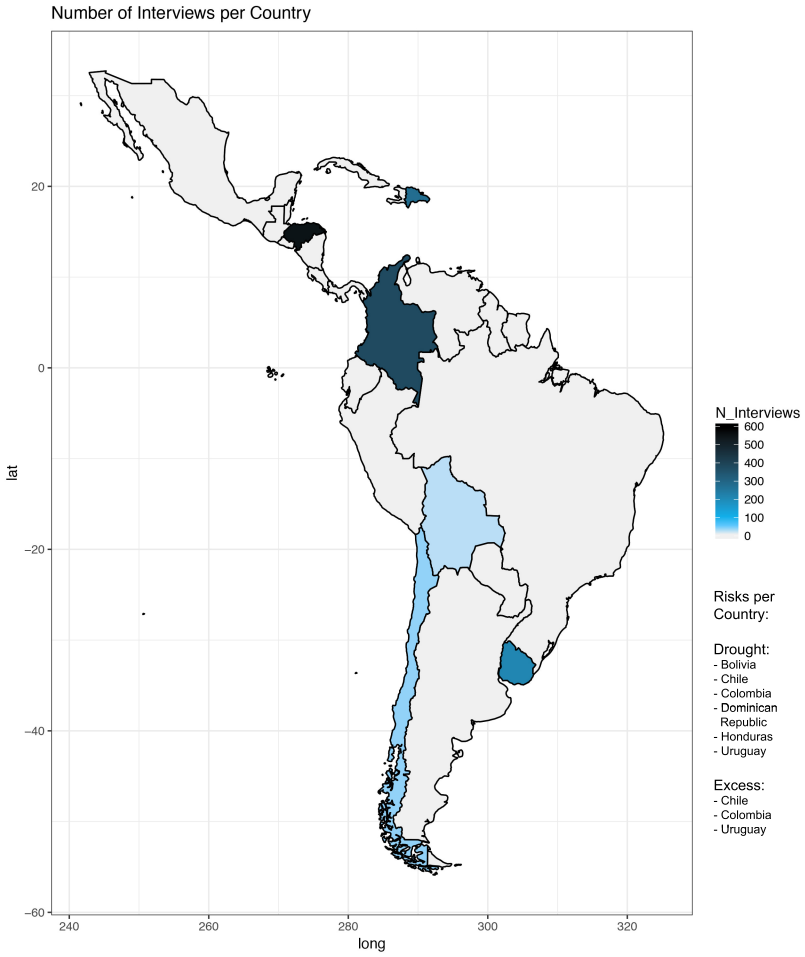
3.1.1 Farmer Bad Years

The first one is designed to understand the performance of the LatAm Dataset in capturing the worst years mentioned by the farmers (information leveraged from other CIF and non-CIF IRI projects, as allowed by project partners and donors). It provides a sense of the performance of the dataset in capturing extreme events. This was performed both for drought and for excess rainfall risks mentioned by the farmers.

There are several reasons why we would expect farmer bad year interviews to have a substantial amount of noise, and perhaps be too noisy to be of use, such as recollection errors and omissions, psychological biases, anchoring, social biases, and economic and political incentives. This is particularly likely in the current work, in which much of the bad year data is through preliminary,

broad interviews. Therefore there are two key utilities in this type of exercise. First, if a data source identifies fewer bad years than other data sources, it is likely that the reported events did occur, since they are evident in other remote sensing data sources. Therefore the data source that did not detect them is likely to be inferior. Second, if a high level of matching is observed, in spite of likely reporting errors in bad years, that is a strong indicator of skill in the remote sensing dataset, as it is performing well even when including reporting noise.

Map 2: Amount of interviews carried out by IRI in the different projects



Analytical process

The first step was to rank the years by the level of precipitation (or climate variable) to identify X number of worst years (in this case X=4). A new binary variable is created which takes the value of 1 if a certain year is one of the X worst years, and 0 otherwise. This new variable is compared with the farmers´ binary bad years to calculate the number of hits and misses. The timing of the year used was also provided by the farmer, when discussing the timing of the risk their production cycle faced. This analysis was carried out not only for the LatAm dataset but also for other satellite datasets[3] as a way to compare the performance of the LatAm dataset with other satellite sources.

[3] CHIRPS, CHIRP, TRMM, EVI, NDVI (EVI and NDVI were just used for the drought risk)

3.1.2 Other sources

The second analysis using the binary approach shows how the LatAm and other datasets are capturing major drought and excess rain events across Latin America. Through a desktop study of different sources,[4] IRI compiled a list of drought and excess rain events across Latin American countries.

Analytical Process

Each of the events identified has a specific timing, so again the datasets were ranked for each year by the level of precipitation (or climate variable), but this time only taking the worst 3 years to see if the event was captured or not by the datasets analyzed.

The main difference between both analysis is that in the first one we had a time series of loss events for the specific critical periods of a production cycle (what would translate into a window in index insurance design) meanwhile in the second analysis we had extreme events at any time of the year. Therefore, the second approach provides a more general insight and indication of the skill of the dataset capturing extreme events, even though not to a specific index insurance design.

3.2 Logit regressions

We test the predictive skill of the dataset for farmer/expert bad years using a logistic regression. This analysis was only carried out for the first type of data described in the section above, because in order to run regression it is necessary to have a timeseries, not only specific events. The model estimates the probability of a binary outcome variable given the variation of a continuous independent variable.

Analytical Process

In this case, the "Bad Years" binary information gathered in the multiple projects was used as the outcome variable, and the precipitation from the LatAm Dataset and others as the explanatory factor. The result of such an exercise is a coefficient that estimates the change of the probability of a year being mentioned as bad, based on an increase in precipitation. This formal statistical method is a more comprehensive approach because it considers the entire precipitation distribution, as opposed to the binary matching that only focuses on a specific set of bad years.

The expected results for the logit regressions are negative coefficients for drought, and positive for excess rainfall. Also, achieving statistical significance in these coefficients would indicate that the product is more adequately reflecting what the farmers experienced on the ground.

[4] World Bank, FAO, NOOA, WFP, emdat.be, mapreport.com.

3.3 Alternate Satellite Sources

As an alternate validation method, the team will compare the physical processes estimated on other types of satellite products as corroborating evidence of the Latam dataset performance and skill for use in index insurance. In doing so, we intend to confirm if the LatAm Dataset reflects the "Bad Years" flagged by other datasets.

The datasources included in this analysis were CHIRPS, CHIRP, TRMM, NDVI and EVI. It is worth mentioning that both NDVI and EVI were only used for the drought analysis given that these datasets don't possess the skill to identify excess rainfall risks due to their intrinsic characteristics. All of these are available at the IRI Data Library (see section 4 for details on each dataset and IRI Data Library links).

The objective of evaluating the LatAm dataset against others, is to evaluate whether the LatAm Dataset is performing better or worse than other datasets available. Other than adding to the validation's robustness, it will inform on the product´s accuracy and potential to be used to trigger payouts of a weather index insurance product and/or monitor seasonal rainfall events. This is a key step that will help the index insurance community better understand where the LatAm dataset is position against other options for design processes.

4 Initial Validation Results

4.1 Binary Matching with Farmer Field Information

We see strong matching between the satellite datasets and the farmer bad years, especially in the countries that IRI participated of index design projects, with several iterations of gathering data in the field. This is the case of Honduras and Uruguay, where we see the highest matching. In Colombia, IRI also carried out an index design project with several instances of gathering farmer data, but we don't see as strong of a relation as in the other two countries mentioned above. The reason why in Colombia we see less hits could be explained by the geographic characteristics of the project. The project was carried out in a mountainous region, where the variations of precipitation in a few kilometers is significant, therefore the minimum resolution[5] of the analysis could explain this lower matching. In fact for the project carried out by IRI the resolution used in the indexes was 25 km^2 (equivalent to 1 CHIRPS pixel). In Chile and Bolivia, we expected lower matching given that only preliminary discussions have been carried out with farmers and local experts. For these two case studies, we would expect that the quality of the bad years gathered would increase as more iteration and more activities to gather data are performed, thus seeing a potentially improved performance of the LatAm dataset, as well as the others.

Overall looking at LatAm in comparison with the other satellite data sources, we see it is competitive, performing slightly better or slightly worse depending on the regions. For example, CHIRPS has a strong performance, with the LatAm dataset being competitive and with better results in some applications. We can see that LatAm is outperformed slightly by CHIRPS in the excess rainfall application in all three countries, as well as in the drought application for Bolivia and Dominican Republic. LatAm is outperforming the other datasets in Colombia and Uruguay for drought and has similar results to CHIRPS in Chile and Honduras. TRMM is not as strong as the other datasets in almost all places, only seems strongest in Colombia for excess rainfall.

Given this analysis the LatAm appears as a competitive dataset to be use for index insurance schemes. For example, prototype indexes developed by IRI for the Colombia case study, Uruguay and in Honduras were based on CHIRPS. In tables 1 and 2 we are seeing that LatAm captures a higher number of bad years than CHIRPS in Colombia and similarly performs in Honduras and Uruguay[6], meaning that it could be an option to consider in an index design process, working through relevant trade-offs such as latency and spatial scale.

Another important factor to have in mind is that through tuning the parameters of the indexes it is possible to improve the performance in capturing bad years. Since index insurance products incorporate specific parameters that allow tuning for local and specific needs, using raw data for validation is less specific to the risk coverage needs.

[5] The minimum resolution of the analysis corresponds to the minimum resolution of the LatAm dataset.
[6] Countries were IRI developed indexes based on CHIRPS

Table 1: Number and percentage of drought hits – 625 Km² resolution

Country	Nº Bad Years	LatAm	CHIRPS	CHIRP	TRMM	NDVI	EVI
		Number of Hits					
Bolivia	8	3	7	4	2	4	1
Chile	6	4	4	5	4	3	3
Colombia	20	9	8	8	9	6	7
Dominican Republic	14	5	6	5	6	5	5
Honduras	16	13	14	10	9	10	9
Uruguay	12	9	9	7	8	9	6

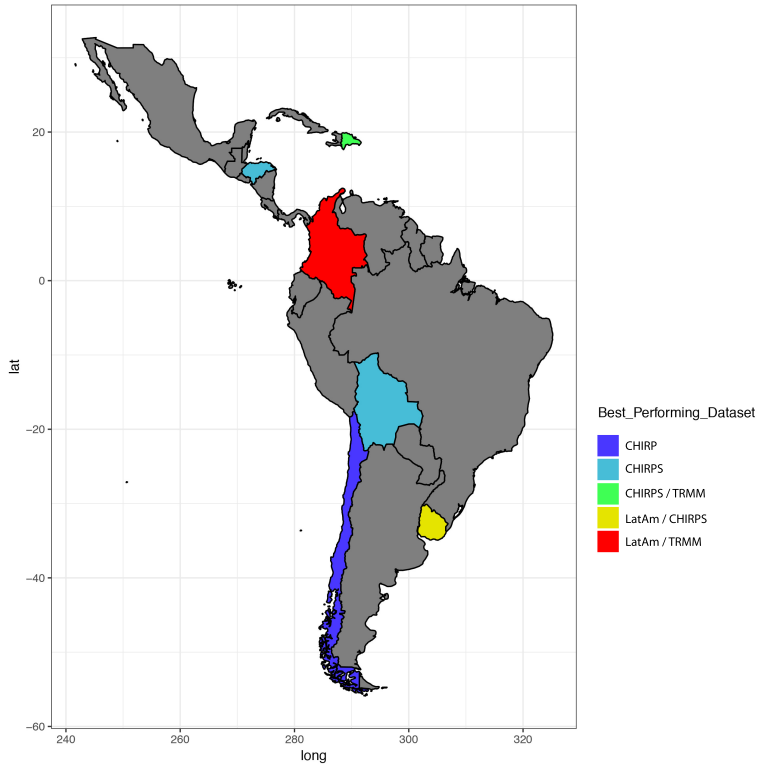
Country	Percentage of Hit					
Bolivia	37.5%	87.5%	50.0%	25.0%	50.0%	12.5%
Chile	66.7%	66.7%	83.3%	66.7%	50.0%	50.0%
Colombia	45.0%	40.0%	40.0%	45.0%	30.0%	35.0%
Dominican Republic	35.7%	42.9%	35.7%	42.9%	35.7%	35.7%
Honduras	81.3%	87.5%	62.5%	56.3%	62.5%	56.3%
Uruguay	75.0%	75.0%	58.3%	66.7%	75.0%	50.0%

Table 2: Heidke Skill Score for drought – 625 Km² resolution

Country	LatAm	CHIRPS	CHIRP	TRMM	NDVI	EVI
Bolivia	0.183	0.837	0.346	0.019	0.346	-0.144
Chile	0.595	0.595	0.798	0.595	0.393	0.393
Colombia	0.316	0.254	0.254	0.316	0.129	0.191
Dominican Republic	0.190	0.280	0.190	0.280	0.190	0.190
Honduras	0.755	0.837	0.510	0.428	0.510	0.428
Uruguay	0.696	0.696	0.494	0.595	0.696	0.393

The first map below displays which dataset possess the best HSS for each country and the second map shows the difference in HSS between the LatAm and CHIRPS datasets in each country for the drought risk (HSS LatAm – HSS CHIRPS).

Map 3: Best HSS performing dataset by Country for drought risk



Map 4: Difference in HSS between LatAm and CHIRPS datasets for drought

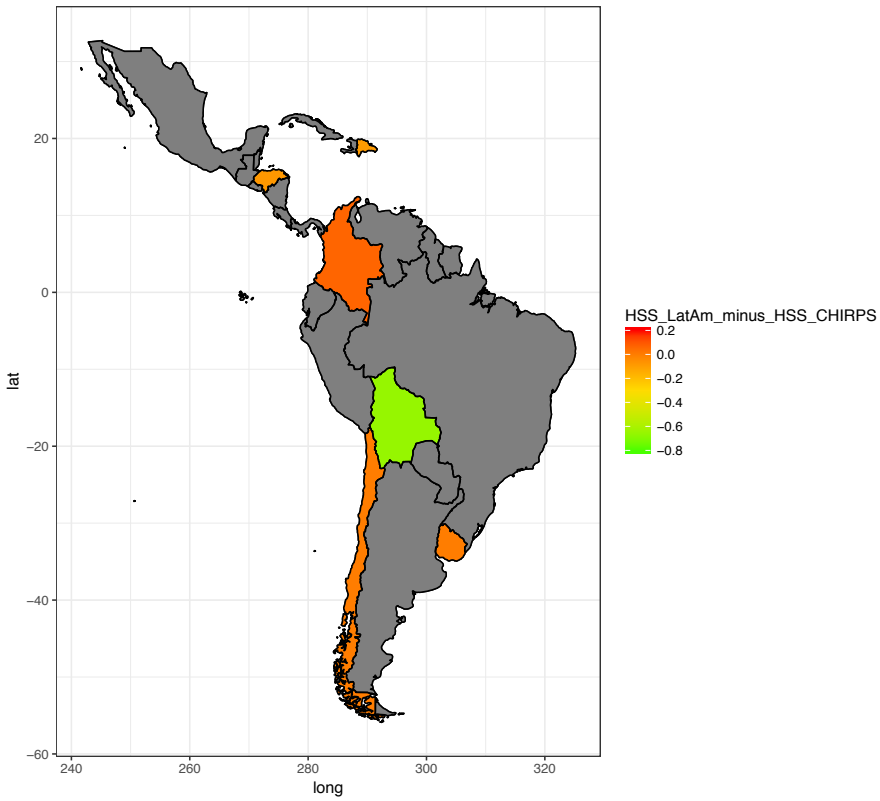


Table 3: Number and percentage of Excess rainfall hits – 625 Km² resolution

Country	Nº Bad Years	LatAm	CHIRPS	CHIRP	TRMM
		Number of Hits			
Chile	7	4	5	4	3
Colombia	29	18	20	14	20
Uruguay	18	13	15	13	10

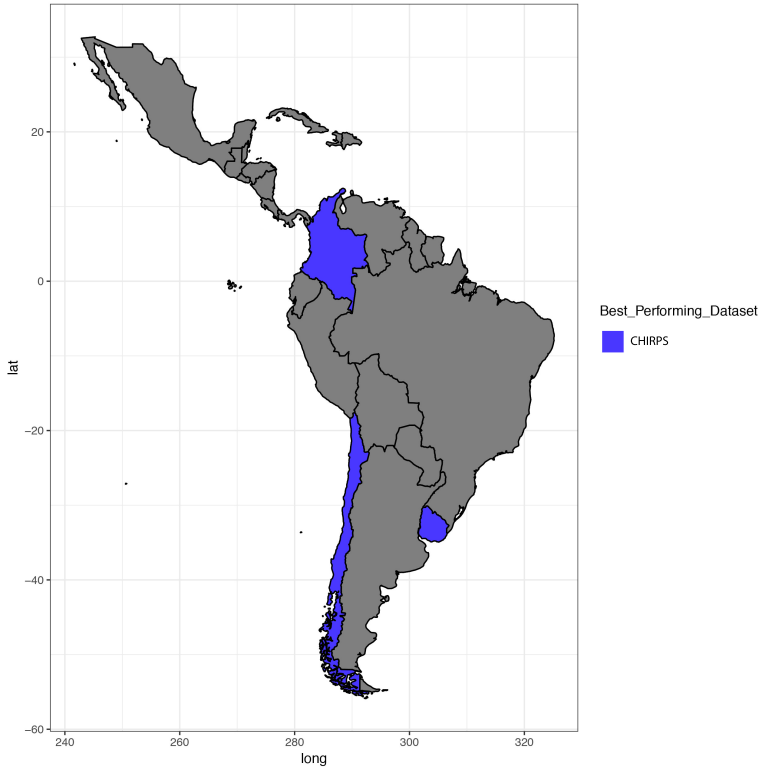
Country	Percentage of Hit			
Chile	57.1%	71.4%	57.1%	42.9%
Colombia	62.1%	69.0%	48.3%	69.0%
Uruguay	72.2%	83.3%	72.2%	55.6%

Table 4: Heidke Skill Score for excess – 625 Km² resolution

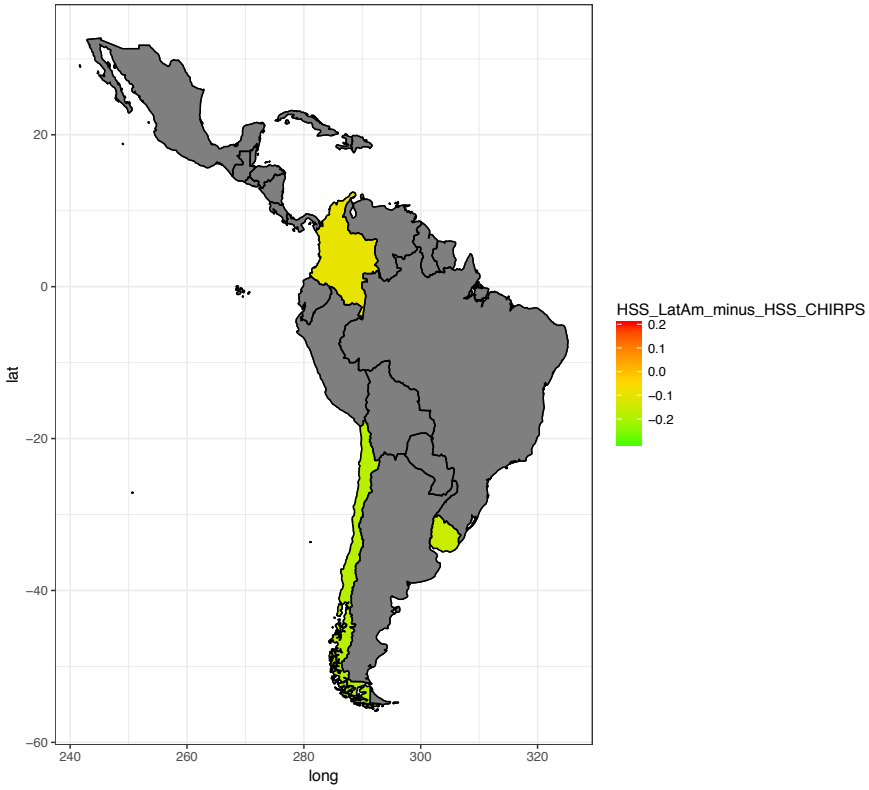
Country	LatAm	CHIRPS	CHIRP	TRMM
Chile	0.468	0.645	0.468	0.113
Colombia	0.481	0.576	0.293	-0.084
Uruguay	0.630	0.778	0.630	-0.037

The first map below displays which dataset possess the best HSS for each country and the second map shows the difference in HSS between the LatAm and CHIRPS datasets in each country for the excess rainfall risk (HSS LatAm – HSS CHIRPS).

Map 5: Best HSS performing dataset by Country for excess rainfall risk



Map 6: Difference in HSS between LatAm and CHIRPS datasets for excess rainfall



4.2 Logit Regression with Farmer Field Information

Logit regressions for drought should yield negative coefficients, meaning that a decrease in precipitation is associated with a higher probability that the year will be identified as bad. It is important to emphasize that the magnitude of the coefficient should not be interpreted directly as a percentage effect on probability. However, it is possible to compare between the coefficients; that is, if one coefficient is greater than another, the effect of said variable will also be greater.

The asterisks represent the statistical significance (or confidence) of the results. In practical terms, if the coefficient has a star, it means that there is a 10% probability that the estimate is incorrect; if it has two stars, the probability is 5%; and if it has a star, this probability is 1%. If there is a coefficient without stars, it means that there is more than a 10% probability that the estimate is erroneous.

It is important to note a few points to explain how we see strong matching in the section above, while not all logit regressions are significant.

First, in most of the case studies the risk in question is present at more than one period over the year, each with two periods: Early and Late. These periods are the result of the vulnerability timings mentioned by the farmers when IRI carried out the field workshops[7]. In other words the two periods are the corresponding timings of the year where the farmers are vulnerable to each risk. Therefore two periods for each dataset were analyzed. Unfortunately, the bad years information gathered usually does not specify which of the two periods of a year was 'bad' for each year mentioned as bad. Looking at the matching rate this issue is not relevant because if in either period the bad year was captured we could assign a match to that year. However for the regressions, this reduces the desired specificity by using either of the periods to predict the bad years. Since we don't have information about which of the bad years correspond to either the Early or Late coverage period, we see an impact on the quality of the fit. To illustrate how the logit regression improves when run specified to one of the periods refer to table 25 in annex 5. These results show a logit regression run for the early period and its corresponding bad years in Uruguay.[8]

Another important consideration is that by tuning the indexes to the specific timing of the risk exposure, as opposed to using a general approach as done in this study, we would expect to see improved fits. As an example, for the Dominican Republic case study we run a logistic regression using the actual NDVI index payouts to predict farmers bad years, obtaining significant and improved results (see table 22 in annex 5) from the general logit regression. We can see that the performance of a regression that takes into account index design specifics can improve highly, which is a an important perspective from which to analyze the results below.

Similar to the binary matching, we expect to obtain better results among the case studies in which we were more involved in gathering loss data on the ground, or those that are at more mature stages, as is the case of Honduras and Uruguay.

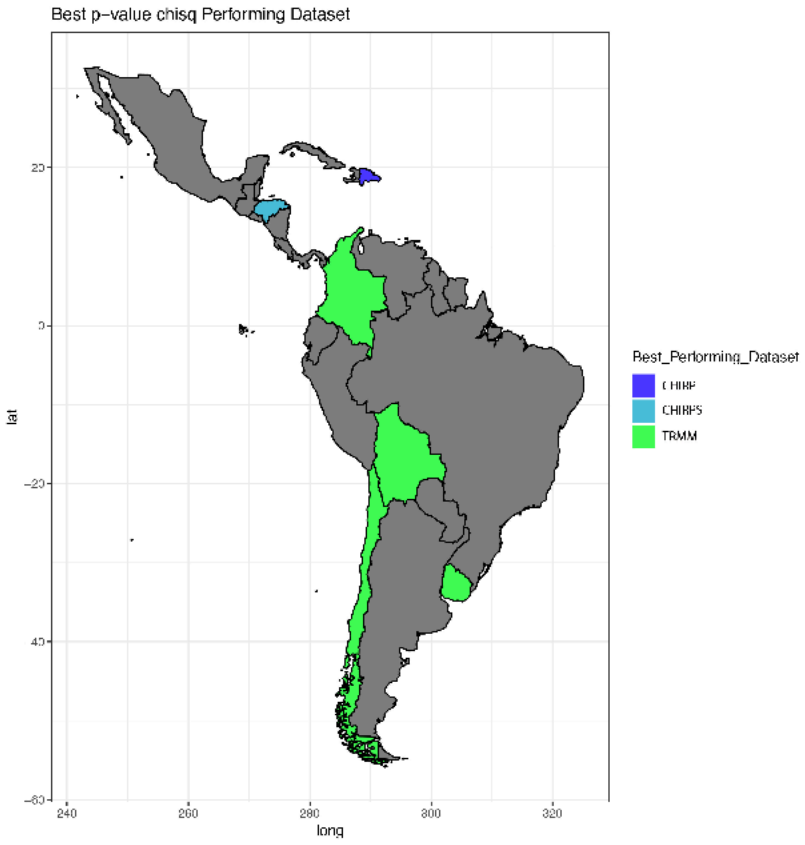
[7] For projects related to crops, the two periods correspond usually to short periods at the start and end of the production season (after planting and before harvesting), while the projects involving dairy productions usually the periods are longer and correspond to seasons of the year (e.g. fall).
[8] In the Uruguay project the bad years gathered were identified for each of the two coverage periods specifically.

Following the Logistic regression a Likelihood Ratio statistics study was performed. Given that the model is binomial, the deviance is approximately $\chi 2$ distributed with n–l degrees of freedom (if the model is correct). Thus we can use the deviance[9] to test whether the model is an adequate fit[10]. The results are shown in the tables and maps below.

Table 5: p-value Chi-squared for drought – 625 Km² resolution

Country	LatAm	CHIRPS	CHIRP	TRMM	NDVI	EVI
Uruguay	0.937	0.847	0.784	0.947	0.870	0.773
Honduras	0.476	0.968	0.372	0.224	0.582	0.733
Colombia	0.583	0.496	0.557	0.666	0.486	0.512
Dominican Republic	0.364	0.350	0.401	0.237	0.355	0.347
Chile	0.704	0.839	0.935	0.947	0.564	0.527
Bolivia	0.210	0.384	0.222	0.731	0.223	0.215

Map 7: Best p-value chi-square performing dataset by Country for drought risk



[9] The deviance measures how close the model comes to perfection. Thus deviance is a measure of goodness of fit.
[10] Faraway, Julian. "Extending the Linear Model with R". 2006

Map 8: Difference in p-value chi-square between LatAm and CHIRPS datasets for drought

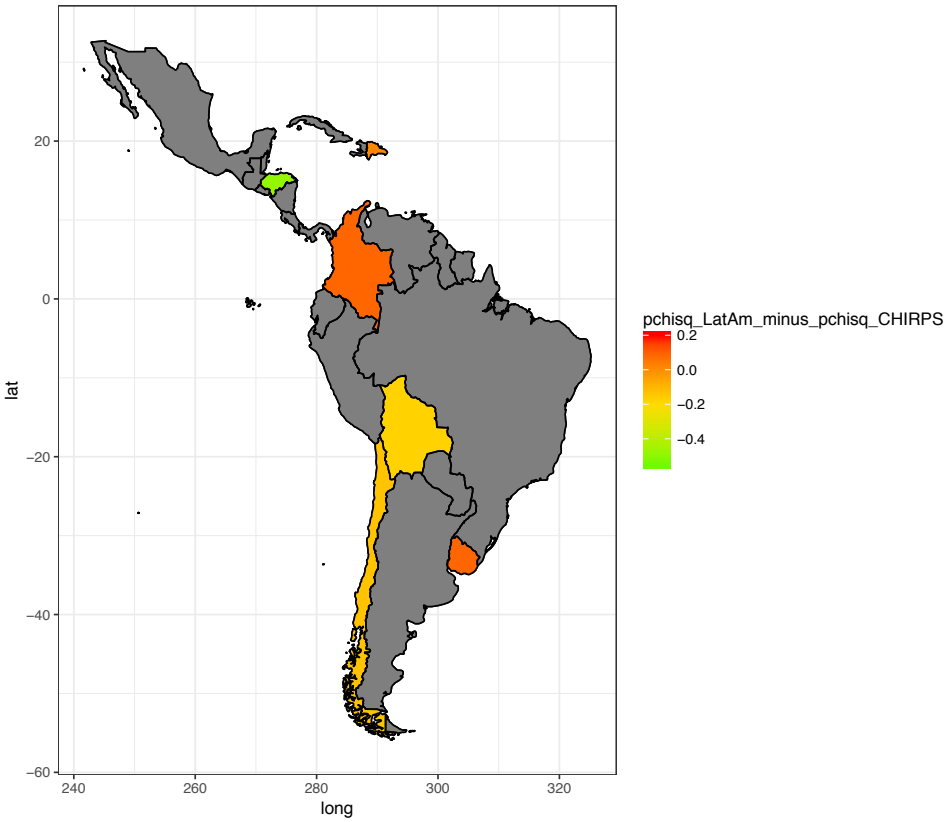
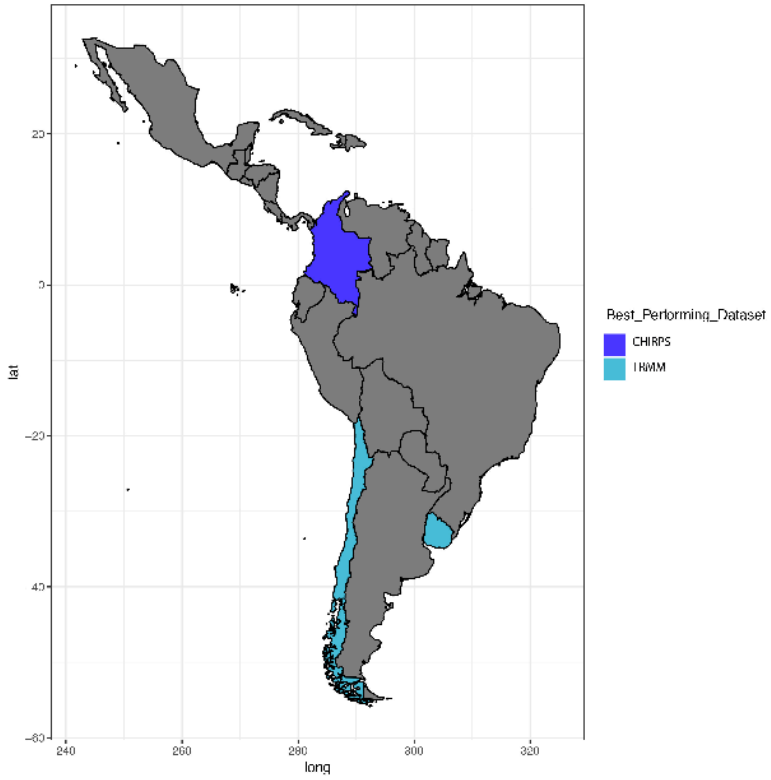


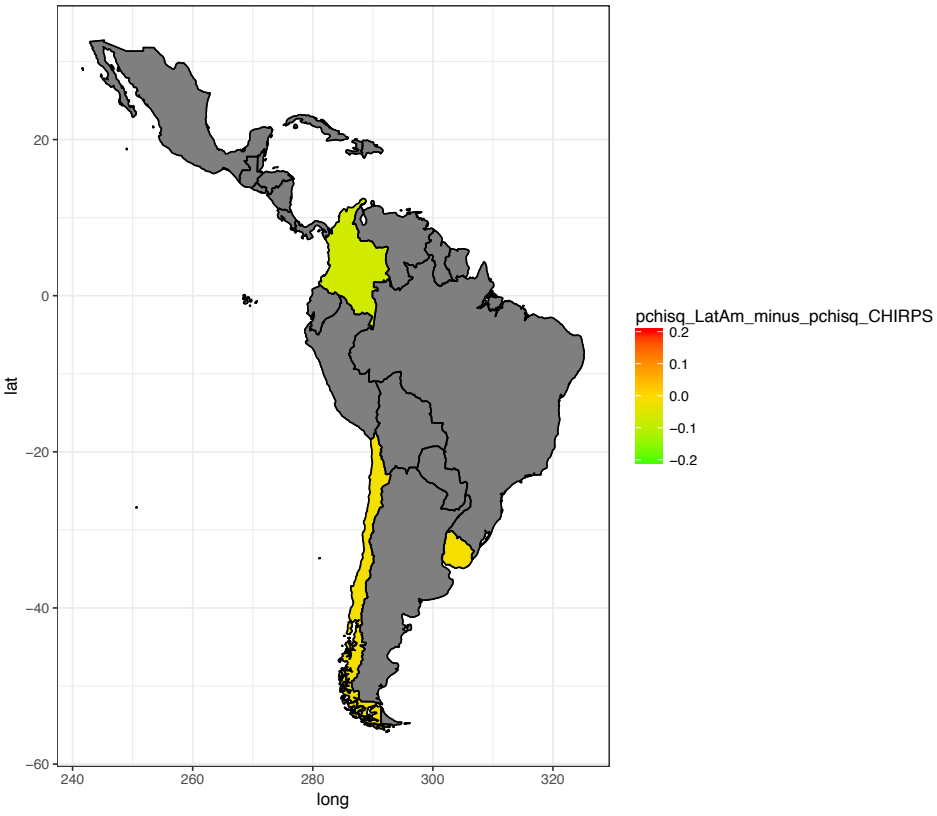
Table 6: p-value Chi-squared for excess – 625 Km² resolution

Country	LatAm	CHIRPS	CHIRP	TRMM
Uruguay	0.273	0.290	0.210	0.339
Colombia	0.095	0.165	0.086	0.048
Chile	0.460	0.479	0.446	0.484

Map 9: Best p-value chi-square performing dataset by Country for excess risk



Map 10: Difference in p-value chi-square between LatAm and CHIRPS datasets for excess



4.3 Binary Matching with Extreme Climate Events

The next two tables show the level of agreement or matching of the satellite datasets and extreme events (floods[11] and droughts) across Latin America. As mentioned the satellite data sources were ranked for each location and the specific timing of the event and a match was considered when the data source had within the 3 worst years (dryer for drought and wetter for flood) the extreme event year considered in the database. Overall we can see LatAm and CHIRPS with the strongest performance among the satellite sources.

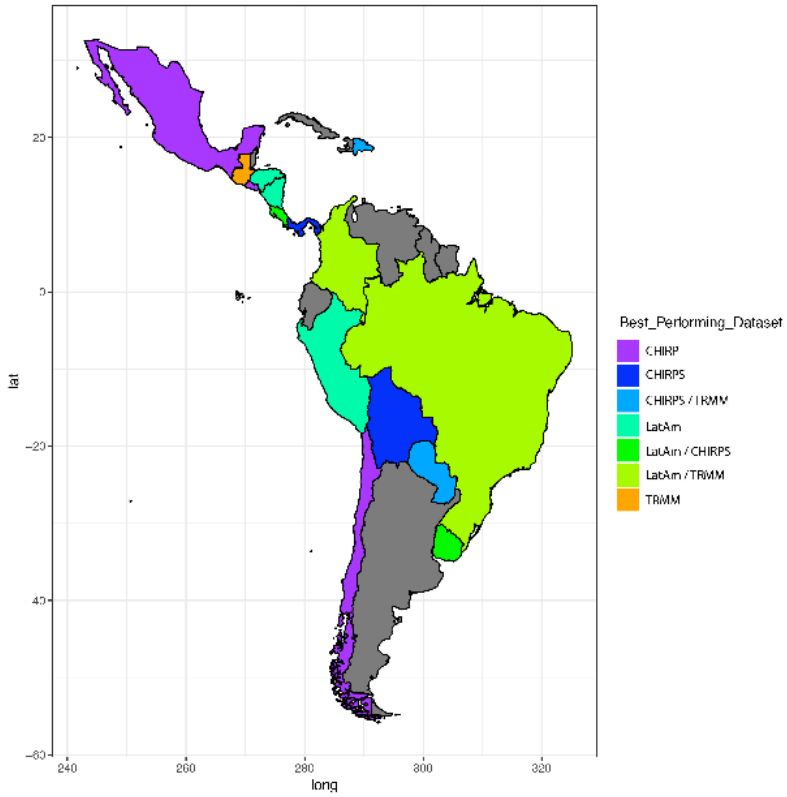
Table 7: Number and Percentage of drought events captured by satellite sources

Country	Nº Bad Years	LatAm	CHIRPS	CHIRP	TRMM
		Number of hits (how many of the "bad years" from the LatAm dataset are capture by other satellite sources)			
Bolivia	10	2	6	3	0
Brazil	8	5	3	2	5
CostaRica	7	6	6	2	0
Ecuador	1	0	0	0	0
ElSalvador	5	1	3	5	4
Guatemala	19	8	5	9	10
Honduras	21	12	5	6	11
Mexico	16	3	6	7	4
Nicaragua	11	5	4	3	0
Panama	5	2	5	1	2
Paraguay	13	4	5	4	5
Peru	3	1	0	0	0

Country	% of bad year matching between LatAm dataset and			
Bolivia	20%	60%	30%	0%
Brazil	63%	38%	25%	63%
CostaRica	86%	86%	29%	0%
Ecuador	0%	0%	0%	0%
ElSalvador	20%	60%	100%	80%
Guatemala	42%	26%	47%	53%
Honduras	57%	24%	29%	52%
Mexico	19%	38%	44%	25%
Nicaragua	45%	36%	27%	0%
Panama	40%	100%	20%	40%
Paraguay	31%	38%	31%	38%
Peru	33%	0%	0%	0%

[11] Flash flood and riverine flood were not considered in this analysis.

Map 11: Best percentage of hit performing dataset by Country for drought risk



Map 12: Difference in percentage of hit between LatAm and CHIRPS datasets for drought

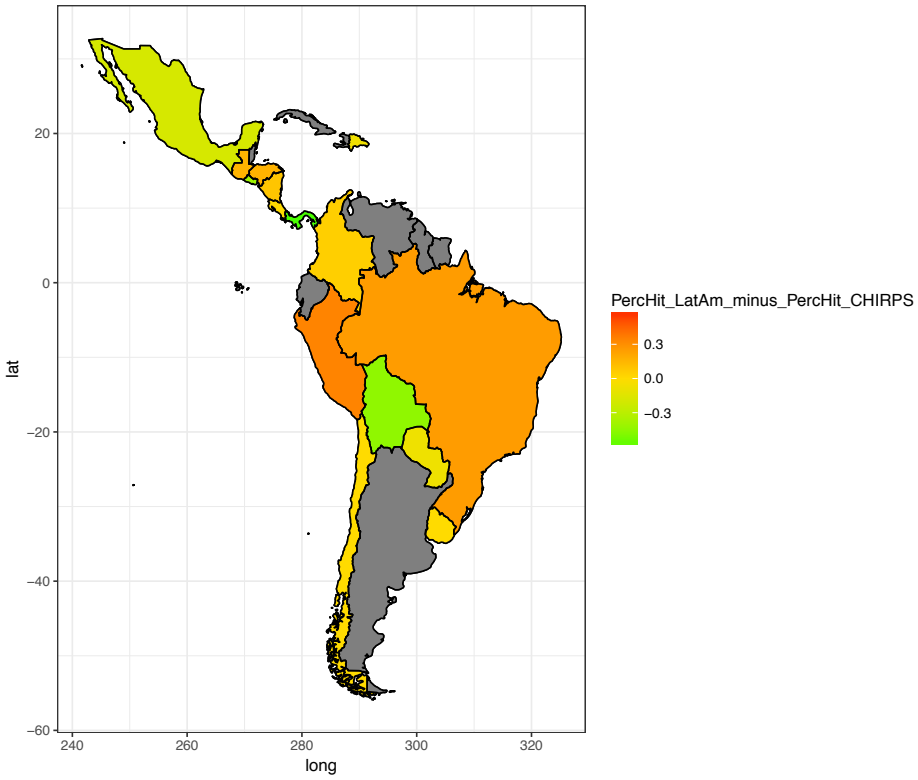
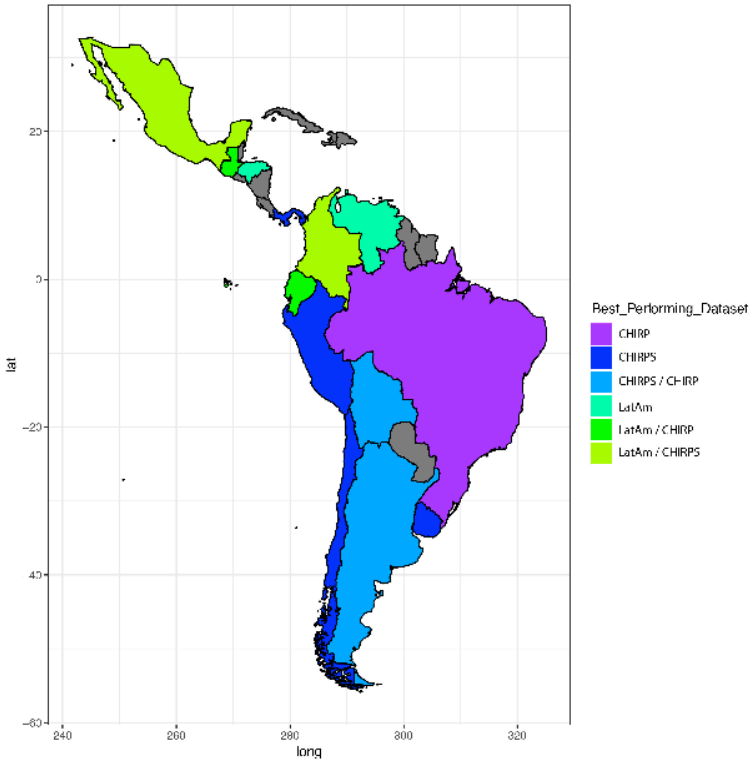


Table 8: Number and Percentage of flood events captured by satellite sources

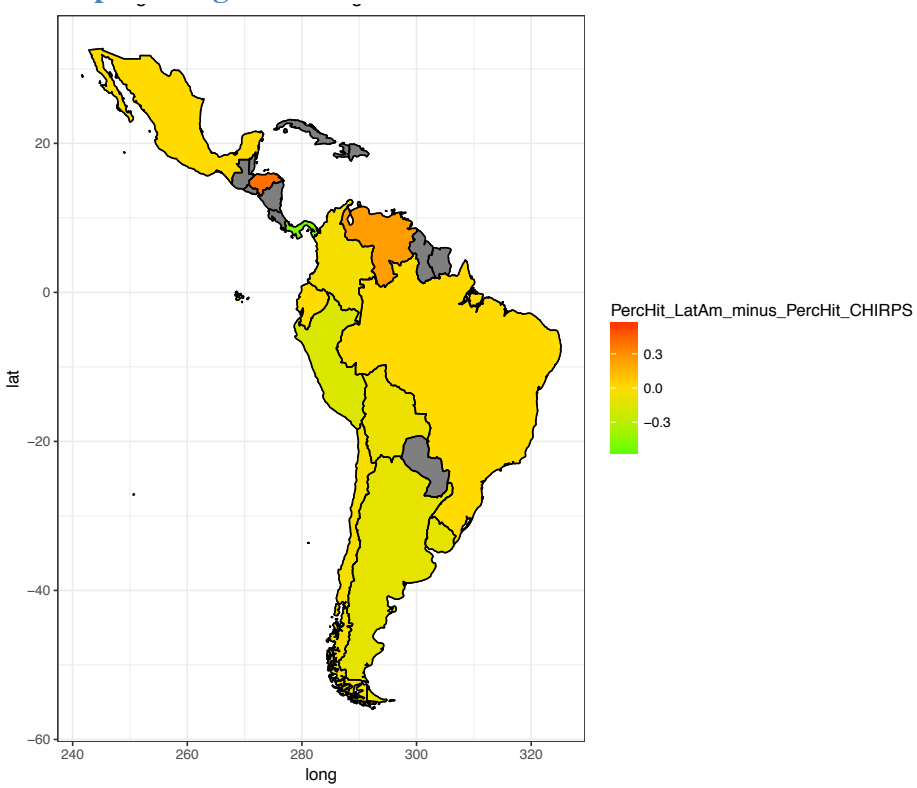
Country	Nº Bad Years	LatAm	CHIRPS	CHIRP	TRMM
		Number of Hits			
Argentina	9	4	5	2	3
Bolivia	12	2	3	2	0
Brazil	8	3	3	3	0
Chile	15	6	7	6	7
Colombia	8	3	3	1	2
Ecuador	3	3	3	1	3
Guatemala	3	2	0	2	0
Honduras	10	5	1	5	1
Mexico	7	2	2	1	2
Panama	2	1	2	0	1
Peru	6	4	5	3	0
Uruguay	17	7	10	0	1
Venezuela	4	4	3	2	0

Country	Percentage of Hit			
Argentina	44%	56%	22%	33%
Bolivia	17%	25%	17%	0%
Brazil	38%	38%	38%	0%
Chile	40%	47%	40%	47%
Colombia	38%	38%	13%	25%
Ecuador	100%	100%	33%	100%
Guatemala	67%	0%	67%	0%
Honduras	50%	10%	50%	10%
Mexico	29%	29%	14%	29%
Panama	50%	100%	0%	50%
Peru	67%	83%	50%	0%
Uruguay	41%	59%	0%	6%
Venezuela	100%	75%	50%	0%

Map 13: Best percentage of hit performing dataset by Country for excess rainfall risk



Map 14: Difference in percentage of hit between LatAm and CHIRPS datasets for excess



4.4 LatAm Data Set Matching with Other Satellite Sources

Table 9: Drought Bad year matching between LatAm dataset and other satellite sources

Country	Nº Bad Years	CHIRPS	CHIRP	TRMM	NDVI	EVI
		Number of hits (how many of the "bad years" from the LatAm dataset are capture by other satellite sources)				
Bolivia	10	6	7	6	4	5
Chile	10	7	5	7	3	3
Colombia	30	19	14	17	10	10
Dominican R.	20	10	12	12	7	8
Honduras	20	14	14	14	9	8
Uruguay	20	15	12	15	10	8

Country	% of bad year matching between LatAm dataset and other satellite sources				
Bolivia	60%	70%	60%	40%	50%
Chile	70%	50%	70%	30%	30%
Colombia	63%	47%	57%	33%	33%
Dominican Republic	50%	60%	60%	35%	40%
Honduras	70%	70%	70%	45%	40%
Uruguay	75%	60%	75%	50%	40%

Table 10: Correlations between LatAm dataset and other satellite sources for each place in the case studies for Drought coverage windows

Region	Country	CHIRPS 1º	CHIRPS 2º	CHIRP 1º	CHIRP 2º	TRMM 1º	TRMM 2º	NDVI 1º	NDVI 2º	EVI 1º	EVI 2º
San Julian	Bolivia	0.47	0.65	0.54	0.52	0.83	0.70	-0.51	0.62	-0.20	0.63
Yapacani	Bolivia	0.66	0.32	0.59	0.53	0.81	0.85	-0.36	0.19	-0.47	0.15
La Union	Chile	0.63	0.89	0.40	0.85	0.84	0.94	0.20	-0.43	0.20	-0.22
Teodoro Sch	Chile	0.86	0.93	0.46	0.83	0.66	0.93	-0.06	-0.48	-0.12	-0.32
Guaimaral	Colombia	0.79	0.62	0.80	0.64	0.75	0.66	-0.36	-0.16	-0.41	-0.16
La Vega	Colombia	0.82	0.47	0.65	0.43	0.70	0.63	-0.01	-0.35	-0.22	0.01
Penoles	Colombia	0.89	0.95	0.58	0.65	0.68	0.75	-0.08	-0.73	0.05	-0.49
Rosas	Colombia	0.84	0.79	0.59	0.72	0.79	0.66	0.11	-0.14	-0.11	0.16
Sotara	Colombia	0.71	0.82	0.59	0.74	0.84	0.87	0.28	0.10	0.30	0.04
Viboral	Colombia	0.85	0.81	0.73	0.59	0.90	0.88	-0.02	-0.39	-0.01	-0.21
Guayubin	Dominican R.	0.59	0.28	0.63	0.66	0.70	0.63	0.14	0.22	0.10	0.18
Partido	Dominican R.	0.70	0.27	0.78	0.64	0.78	0.58	0.06	0.15	0.04	0.10
Sabaneta	Dominican R.	0.59	0.10	0.70	0.44	0.78	0.64	0.18	-0.14	0.13	-0.15
Villa Vazquez	Dominican R.	0.62	0.36	0.65	0.68	0.71	0.60	0.12	0.19	0.09	0.17
El Barro	Honduras	0.70	0.81	0.68	0.72	0.61	0.87	0.59	0.01	0.47	0.26
El Plan	Honduras	0.69	0.81	0.72	0.72	0.75	0.87	0.45	0.02	0.41	0.26
Los Limones	Honduras	0.64	0.82	0.66	0.74	0.63	0.82	0.45	0.24	0.40	0.15
Mesias	Honduras	0.64	0.83	0.66	0.70	0.63	0.81	0.45	0.15	0.42	0.35
Canelones	Uruguay	0.96	0.87	0.83	0.55	0.98	0.90	0.35	0.16	0.05	-0.05
Carmelo	Uruguay	0.90	0.92	0.75	0.60	0.97	0.97	0.57	0.41	-0.19	0.43
San Jose	Uruguay	0.93	0.91	0.80	0.60	0.98	0.95	0.40	0.16	0.08	-0.23
Tres Bocas	Uruguay	0.98	0.92	0.92	0.60	0.95	0.96	0.72	0.24	0.47	0.16

A 2D visualization of the correlation "landscape" performing a Principal Component Analysis (PCA), retaining 2 principal components and projecting the position of the dataset locations on the plane defined by the first 2 loadings is shown below.

Figure 1: PCA landscape for drought

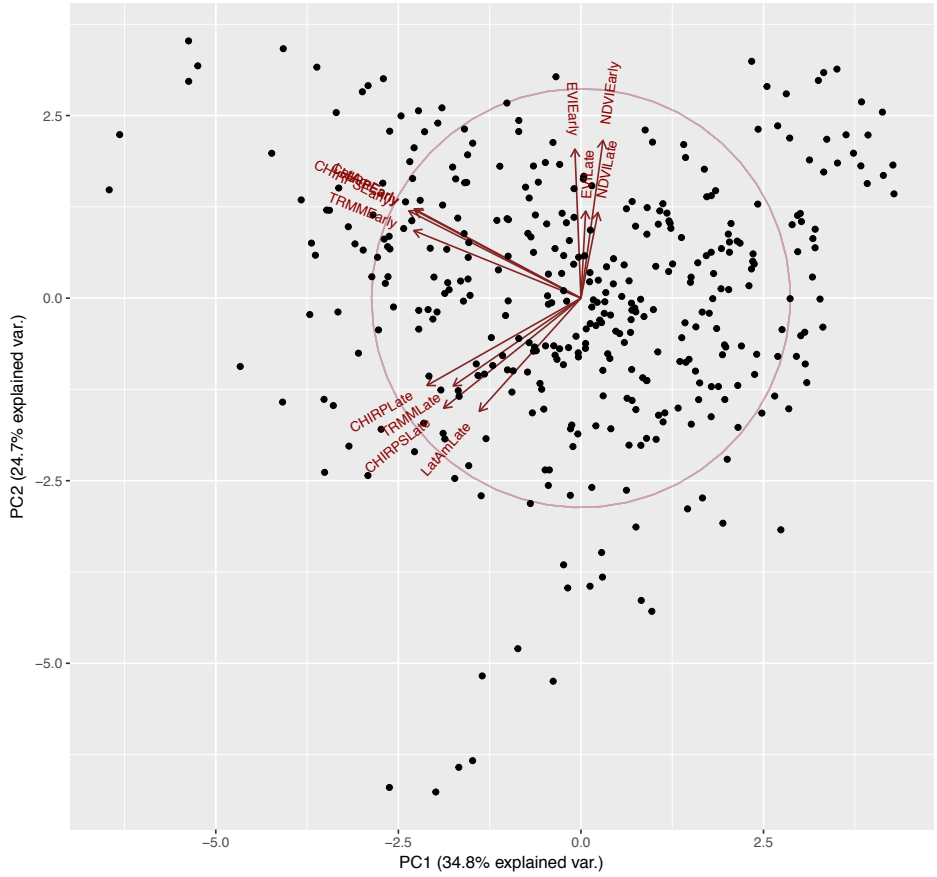


Table 11: Excess rainfall Bad year matching between LatAm dataset and other satellite sources

Country	Nº Bad Years	CHIRPS	CHIRP	TRMM
		Number of hits (how many of the "bad years" from the LatAm dataset are capture by other		
Chile	10	7	5	7
Colombia	30	20	15	17
Uruguay	20	16	12	15

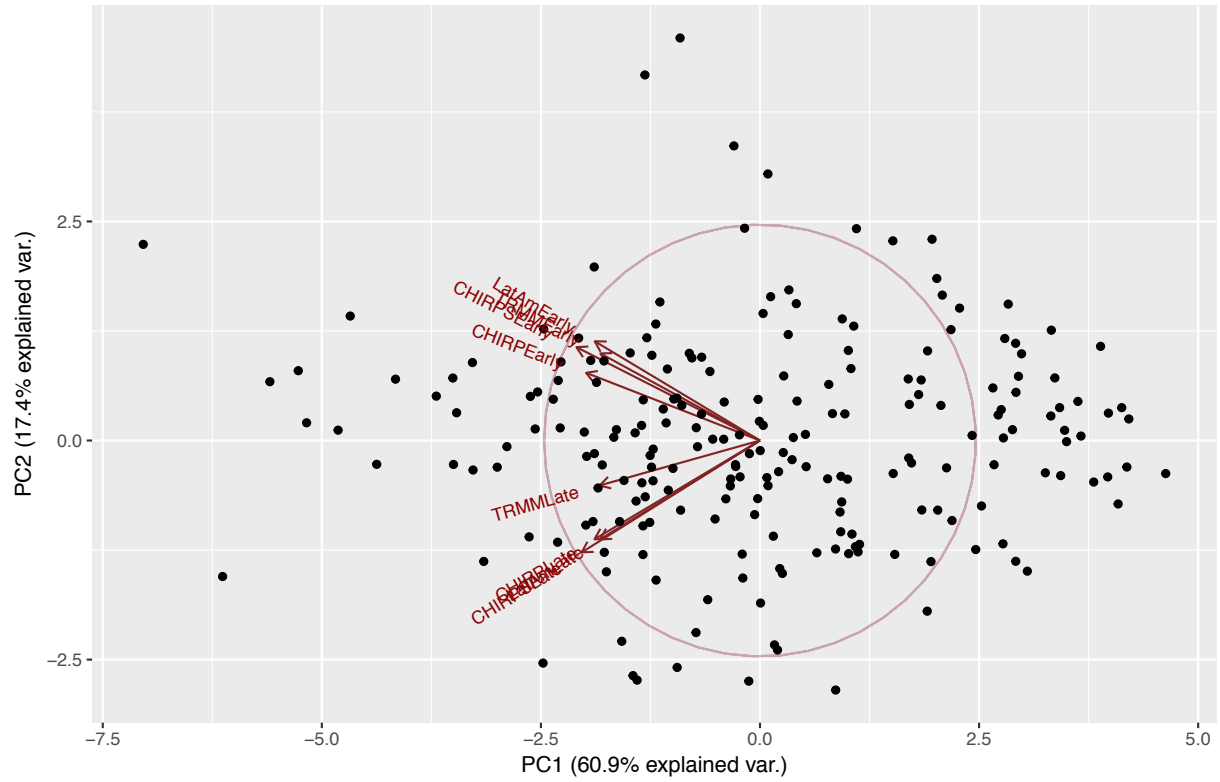
Country	% of bad year matching between LatAm dataset and other satellite sources		
Chile	70%	50%	70%
Colombia	67%	50%	57%
Uruguay	80%	60%	75%

Table 12: Correlations between LatAm dataset and other satellite sources for each place in the case studies for Excess rainfall coverage windows

Region	Country	CHIRPS 1º	CHIRPS 2º	CHIRP 1º	CHIRP 2º	TRMM 1º	TRMM 2º
La Union	Chile	0.86	0.20	0.66	0.17	0.84	0.69
Teodoro Schmidt	Chile	0.90	0.64	0.57	0.33	0.83	0.89
Guaimaral	Colombia	0.88	0.76	0.82	0.69	0.90	0.59
La Vega	Colombia	0.62	0.39	0.21	-0.04	0.67	0.67
Penoles	Colombia	0.29	0.18	-0.23	-0.59	0.77	0.60
Rosas	Colombia	0.68	0.37	0.25	0.25	0.81	0.58
Sotara	Colombia	0.23	0.24	0.07	0.06	0.79	0.90
Viboral	Colombia	0.84	0.79	0.81	0.63	0.87	0.80
Canelones	Uruguay	0.97	0.98	0.85	0.95	0.98	0.99
Carmelo	Uruguay	0.96	0.95	0.75	0.76	0.97	0.94
San Jose	Uruguay	0.86	0.93	0.81	0.79	0.97	0.92
Tres Bocas	Uruguay	0.97	0.93	0.85	0.64	0.94	0.94

A 2D visualization of the correlation "landscape" performing a Principal Component Analysis (PCA), retaining 2 principal components and projecting the position of the dataset locations on the plane defined by the first 2 loadings is shown below.

Figure 2: PCA landscape for excess



5 Annex 1
Case Studies for Validation Study

This section explores six cases in detail: Colombia, Uruguay, Dominican Republic, Honduras, Bolivia and Chile. All of them are projects in which IRI was directly involved in the collection of perception data, through a methodology of systematic interviews and discussions specifically designed to gather data, as accurate as possible, from groups and farmers in the field (for more details on the methodology refer to Annex 2). The result is a set of binary datasets indicating the worst perceived years for production in each region. It is important to note that in Bolivia and Chile the data gathered is preliminary, thus less consolidated than the others.

This section also presents a list of the drought and excess rainfall events gathered from different sources, such as World Bank, FAO, NOOA, WFP, emdat.be, mapreport.com.

It is worth noting that although each of these projects is dedicated to the use of its data for academic and scholarly purposes, we have to follow the appropriate protocols for access to data. Partners and donors from other non-CIF efforts were contacted ahead of this study.

5.1 Colombia

IRI collaborated with PlaNet Guarantee and Nespresso to assess the potential for developing crop index insurance for coffee farmers in Colombia. The efforts carried-out included gathering feedback from coffee farmers in the field to develop an academic prototype rainfall index. Such an exercise gave IRI the opportunity to gather systematic information about the farmers´ perception regarding which had been the worst years for their production because of climate impacts.

The resulting database consists of a list of "Bad Years" coming from interviews and discussions held in October 2015 with members from different communities of coffee producers. The data gathered covers from 1985 to 2015, and focuses on the coffee production cycles. The data contains information on drought and excess rainfall events, both being relevant hazards for the coffee production. This is particularly convenient for the current study as it allows to differentiate the performance of the product for the two types of hazards.

Table 13: Data Characteristics Colombia

Hazards	Excess Rainfall and Drought
Period	1985 - 2015
Product	Coffee
Number of Regions	6
Production Windows	1 or 2 depending on the region

5.2 Dominican Republic

IRI carried out a performance evaluation of an index insurance product developed for dairy producers in the Dominican Republic for the USAID CRII project. The exercise consisted on gathering farmers´ perceptions data on "Bad Years" through interviews and participatory discussions with sub-groups of producers within the communities. The workshops were held in four different municipalities during 2013 and 2014.

The database contains a list of the worst drought years in the dairy farmers´ experiences to date. The data was collected in the Municipalities of Guayubin, Sabaneta, Partido and Villa Vázquez, all of them located on the northwest of the Dominican Republic. It covers a period from 2001 to 2015, and is divided into two usually-dry windows: January through April; and July through September.

Table 14: Data Characteristics Dominican Republic

Hazards	Drought
Period	2001 - 2015
Product	Dairy
Number of Regions	4
Production Windows	2

5.3 Honduras

IRI in collaboration the Ministry of Agriculture and Livestock (SAG) and Universidad Zamorano, supported by CCAFS, coordinated an index insurance project for maize and beans in Honduras. IRI participated in the design of the index, and performed a series of validations studies on the field including financial games to estimate the demand, and systematic interviews and discussions with more than 100 farmers. As a result, a dataset was put together which allowed to calibrate the index for its future use.

The data consists on a list of "Bad Years" coming from the discussions held with maize and beans farmers on the field. The workshops were held in 2014, and covered a period from 1981 to 2014. The data was gathered in four communities in the El Paraíso department: El Plan, El Barro, and Santa María and San Francisco. All the data is divided into two production seasons: primera and postrera.

Table 15: Data Characteristics Honduras

Hazards	Drought
Period	1981 - 2015
Product	Maize and Beans
Number of Regions	4
Production Windows	2

5.4 Uruguay

IRI in collaboration with the Ministry of Livestock, Agriculture and Fishery (MGAP), and the National Institute of Milk (INALE), worked in four communities of dairy producers in Uruguay, to identify the climate risks and the potential for index insurance in the sector, integrating the farmers´ perception in the design of the indexes through participatory processes. Such an exercise resulted in the collection of "Bad Year" data in the region.

The database contains what the farmers perceive as "Bad Years" for milk production regarding droughts and prolonged excess rainfall. It was gathered during two workshops held on May 2015 and May 2016, in four departments with high density of milk producers: Río Negro, located on the western side of Uruguay; Colonia on the southwestern border of the country; and San José and Canelones on the south.

The data covers a period from 1999 to 2016, and is divided into four seasons. This is relevant because it will allow the team to validate the performance of the LatAm Datset capturing events at different times of the year.

Table 16: Data Characteristics Uruguay

Hazards	Prolonged Excess Rainfall and Drought
Period	1999 - 2016
Product	Milk
Number of Regions	4
Production Windows	4

5.5 Bolivia

Though a CIF funded project, IRI carried out a Feasibility Study for agricultural insurance in Bolivia. During the interviews and meetings conducted with local experts and agricultural institutions, IRI's team was able to discuss preliminary "Bad Years" for 2 different production regions near Santa Cruz. It is important to note that given this project was a feasibility study the "Bad Years" information has not been validated through follow up farmer meetings

5.6 Chile

IRI in collaboration the Ministry of Agriculture, through its AgroSeguros division, is actively carrying out an index design project for dairy prodution in the Region de Los Rios. As of now, preliminary meetings with farmer have taken place where initial "Bad Years" and climate risk information for 2 areas were discussed. Through this year, IRI will perform workshops with producers from where more field information will be gathered. If this information is available before the final validation run, it will be included in the analysis.

5.7 Climate extreme events in Latin American Countries

The tables below shows the extreme climate events gathered by the IRI team from a variety of sources, including World Bank, FAO, NOOA, WFP, emdat.be and mapreport.com.

Table 17: Drought events in Latin American Countries

Location	Country	Long	Lat	Start Dekad	Final Dekad	Event Year
Amazonas1	Brazil	-66.0	-3.3	28	30	2005
Amazonas2	Brazil	-66.0	-3.3	28	30	2010
MinasGerais	Brazil	-44.7	-18.4	25	36	2002
MinasGerais2	Brazil	-44.7	-18.4	28	36	2007
Parana	Brazil	-52.1	-25.1	19	21	2015
Ceara	Brazil	-39.4	-5.4	1	36	2007
Maranhao	Brazil	-45.4	-4.8	1	36	2007
Tocantins	Brazil	-48.4	-10.0	28	36	2007
Neembucu	Paraguay	-57.9	-26.9	1	33	2009
Neembucu2	Paraguay	-57.9	-26.9	1	1	2012
AltoParaguay	Paraguay	-59.6	-20.0	29	36	2005
AltoParaguay2	Paraguay	-59.6	-20.0	25	36	2008
AltoParaguay3	Paraguay	-59.6	-20.0	1	33	2009
AltoParaguay5	Paraguay	-59.6	-20.0	19	24	2016
Boqueron	Paraguay	-61.1	-21.6	29	36	2005
Boqueron2	Paraguay	-61.1	-21.6	1	33	2009
Boqueron4	Paraguay	-61.1	-21.6	19	24	2016
PresidenteHayes	Paraguay	-58.9	-23.2	29	36	2005
PresidenteHayes2	Paraguay	-58.9	-23.2	25	36	2008
PresidenteHayes3	Paraguay	-58.9	-23.2	1	33	2009
PresidenteHayes5	Paraguay	-58.9	-23.2	19	24	2016
Ucayali	Peru	-73.2	-9.7	7	36	2002
Cajamarca	Peru	-78.6	-7.0	1	36	2004
Arequipa	Peru	-71.7	-16.3	2	4	2006
SantaCruz3	Bolivia	-63.3	-17.7	16	27	2010
SantaCruz4	Bolivia	-63.3	-17.7	1	27	2013
SantaCruz5	Bolivia	-63.3	-17.7	28	30	2014
Beni2	Bolivia	-65.2	-14.2	1	27	2013
Beni3	Bolivia	-65.2	-14.2	7	34	2016
Chuquisaca3	Bolivia	-64.3	-19.9	16	27	2010
Chiquisaca4	Bolivia	-64.3	-19.9	1	27	2013

Location	Country	Long	Lat	Start Dekad	Final Dekad	Event Year
Oruro2	Bolivia	-67.2	-17.8	7	34	2016
Cochamba2	Bolivia	-66.3	-17.3	28	36	2013
Cochamba3	Bolivia	-66.3	-17.3	7	34	2016
Loja	Ecuador	-79.3	-3.9	4	16	2013
BajaCalifornia	Mexico	-115.4	31.0	10	36	2002
Coahuila	Mexico	-101.8	27.2	10	36	2002
Tamaulipas	Mexico	-99.0	24.4	10	36	2002
Veracruz	Mexico	-96.3	19.3	10	36	2002
Sinaloa	Mexico	-107.6	25.3	10	36	2002
Tlaxcala	Mexico	-98.4	19.4	10	36	2002
Sonora	Mexico	-110.5	29.4	10	36	2002
SanLuisPotosi	Mexico	-101.1	22.3	10	36	2002
NuevoLeon	Mexico	-100.1	25.7	25	36	2011
Guanajuato	Mexico	-101.4	21.1	25	36	2011
Sonora2	Mexico	-110.5	29.4	25	36	2011
Chihuahua	Mexico	-106.2	28.8	25	36	2011
Coahuila2	Mexico	-101.8	27.2	25	36	2011
Durango	Mexico	-104.8	24.7	25	36	2011
Zacatecas	Mexico	-102.7	22.9	25	36	2011
SanLuisPotosi2	Mexico	-101.1	22.3	25	36	2011
BajaVerapaz2	Guatemala	-90.5	15.2	7	36	2009
ElProgreso2	Guatemala	-90.2	15.0	7	36	2009
Zacapa2	Guatemala	-89.7	15.1	7	36	2009
Jalapa2	Guatemala	-90.0	14.7	7	36	2009
Chiquimula2	Guatemala	-89.7	14.9	7	36	2009
SantaRosa2	Guatemala	-90.5	14.3	7	36	2009
Jutiapa2	Guatemala	-90.0	14.3	7	36	2009
Quiche2	Guatemala	-91.1	15.6	7	36	2009
Chiquimula3	Guatemala	-89.7	14.9	25	36	2001
Jalapa3	Guatemala	-90.0	14.7	16	36	2012
ElProgreso3	Guatemala	-90.2	15.0	16	36	2012
Zacapa3	Guatemala	-89.7	15.1	16	36	2012
SantaRosa3	Guatemala	-90.5	14.3	16	36	2012
Sacatepequez	Guatemala	-90.8	14.7	16	36	2012
Solola	Guatemala	-91.3	14.9	16	36	2012
Chimaltenango	Guatemala	-90.9	14.8	16	36	2012
BajaVerapaz3	Guatemala	-90.5	15.2	16	36	2012

Location	Country	Long	Lat	Start Dekad	Final Dekad	Event Year
Quiche3	Guatemala	-91.1	15.6	16	36	2012
Huehuetenango2	Guatemala	-91.6	15.4	16	36	2012
Usulutan	ElSalvador	-88.7	13.6	13	36	2001
LaUnion	ElSalvador	-88.0	13.5	13	36	2001
SanMiguel	ElSalvador	-88.3	13.6	13	36	2001
Morazan	ElSalvador	-88.2	13.9	13	36	2001
AllDepartments	ElSalvador	-89.0	13.9	1	36	2009
FranciscoMorazan	Honduras	-87.2	14.6	19	36	2000
FranciscoMorazan2	Honduras	-87.2	14.6	19	36	2001
Choluteca	Honduras	-87.3	13.4	19	36	2001
Paraiso	Honduras	-86.5	14.2	19	36	2001
LaPaz1	Honduras	-88.1	14.1	19	36	2001
Comayagua	Honduras	-87.7	14.7	19	36	2001
Intibuca	Honduras	-88.4	14.5	19	36	2001
Yoro	Honduras	-87.2	15.3	19	36	2001
Lempira	Honduras	-88.7	14.3	19	36	2001
Valle	Honduras	-87.7	13.7	19	36	2001
Paraiso2	Honduras	-86.5	14.2	1	36	2002
Choluteca2	Honduras	-87.3	13.4	1	36	2002
AllDepartments_Hon	Honduras	-86.4	15.3	19	36	2004
Choluteca4	Honduras	-87.3	13.4	25	27	2012
FranciscoMorazan4	Honduras	-87.2	14.6	25	27	2012
GraciasADios	Honduras	-84.7	15.5	25	27	2012
LaPaz2	Honduras	-88.1	14.1	25	27	2012
Olancho	Honduras	-85.9	14.9	25	27	2012
Paraiso4	Honduras	-86.5	14.2	25	27	2012
Valle3	Honduras	-87.7	13.7	25	27	2012
Alldepts_Hon	Honduras	-86.4	15.3	10	19	2015
AtlanticoNorte	Nicaragua	-84.1	14.0	19	36	2000
Jinotega	Nicaragua	-86.1	13.2	19	36	2000
NuevaSegovia	Nicaragua	-86.1	13.9	19	36	2000
NuevaSegovia2	Nicaragua	-86.1	13.9	22	36	2001
Madriz	Nicaragua	-86.6	13.6	22	36	2001
Esteli	Nicaragua	-86.5	13.2	22	36	2001
Leon	Nicaragua	-87.0	12.6	22	36	2001
Chinandega	Nicaragua	-87.2	12.7	22	36	2001
Carazo	Nicaragua	-86.3	11.8	22	36	2001

Location	Country	Long	Lat	Start Dekad	Final Dekad	Event Year
Boaco	Nicaragua	-85.8	12.6	22	36	2001
Matagalpa	Nicaragua	-86.0	13.0	22	36	2001
Alajuela	CostaRica	-84.3	10.1	22	24	2015
Cartago	CostaRica	-84.0	10.0	22	24	2015
Guanacaste	CostaRica	-85.6	10.7	22	24	2015
Heredia	CostaRica	-84.2	10.1	22	24	2015
Limon	CostaRica	-83.2	10.1	22	24	2015
Puntarenas	CostaRica	-84.9	10.1	22	24	2015
SanJose	CostaRica	-84.2	10.0	22	24	2015
Cocle	Panama	-80.5	8.7	1	15	2013
Veraguas	Panama	-81.2	8.2	1	15	2013
LosSantos	Panama	-80.5	7.7	1	15	2013
Herrera	Panama	-80.8	7.9	1	15	2013
AllDeptsPan	Panama	-80.9	8.7	22	24	2015

Table 18: Excess Rainfall events in Latin American Countries

Location	Country	Long	Lat	Start Dekad	Final Dekad	Event Year
Amazonas	Brazil	-66.0	-3.3	9	9	2004
RioGrandeDoSul	Brazil	-51.3	-29.9	32	32	2008
RioGrandeDoSul2	Brazil	-51.3	-29.9	19	21	2015
SantaCatarina	Brazil	-50.3	-27.1	9	9	2004
SantaCatarina2	Brazil	-50.3	-27.1	19	21	2015
RioDeJanairo	Brazil	-43.3	-22.8	10	12	2011
RioDeJanairo2	Brazil	-43.3	-22.8	35	35	2013
SaoPaolo	Brazil	-46.8	-23.4	1	2	2016
DistritoCapital	Venezuela	-67.1	10.6	28	30	2000
Monagas	Venezuela	-63.1	9.4	33	34	2016
Miranda	Venezuela	10.1	10.4	33	34	2016
NuevaEsparta	Venezuela	-64.0	11.1	33	34	2016
BuenosAires	Argentina	-58.5	-34.5	14	14	2000
BuenosAires2	Argentina	-58.5	-34.5	31	32	2000
BuenosAires3	Argentina	-58.5	-34.5	9	9	2001
BuenosAires4	Argentina	-58.5	-34.5	30	31	2014
RioNegro	Argentina	-63.1	-40.7	19	21	2001

Location	Country	Long	Lat	Start Dekad	Final Dekad	Event Year
SantaFe	Argentina	-60.8	-31.5	14	14	2000
SantaFe2	Argentina	-60.8	-31.5	31	32	2000
SantaFe3	Argentina	-60.8	-31.5	32	32	2000
EscalantePatagonia	Argentina	-67.8	-45.3	9	10	2001
Arequipa	Peru	-71.7	-16.3	5	6	2016
Arequipa2	Peru	-71.7	-16.3	2	4	2017
Ancash	Peru	-77.7	-9.2	8	8	2017
Tumbes	Peru	-80.5	-3.4	8	8	2017
Junin	Peru	-75.5	-11.2	5	6	2016
Lima	Peru	-77.2	-11.9	2	4	2017
Tarija	Bolivia	-64.8	-21.4	4	4	2000
LaPaz	Bolivia	-68.2	-16.4	4	4	2000
LaPaz2	Bolivia	-68.2	-16.4	6	7	2016
LaPaz3	Bolivia	-68.2	-16.4	33	34	2016
Cochamba	Bolivia	-66.3	-17.3	4	4	2000
Cochamba2	Bolivia	-66.3	-17.3	6	7	2016
Cochamba3	Bolivia	-66.3	-17.3	33	34	2016
Pando	Bolivia	-66.0	11.6	5	6	2015
Oruro	Bolivia	-67.2	-17.8	4	4	2000
Potosi	Bolivia	-65.9	-19.4	4	4	2000
Chiquisaca	Bolivia	-64.3	-19.9	4	4	2000
SantaCruz	Bolivia	-63.3	-17.7	33	34	2016
Carchi	Ecuador	-78.0	0.6	15	15	2000
Imbabura	Ecuador	-78.2	0.5	15	15	2000
ElOro	Ecuador	-80.1	-3.1	8	8	2002
Aisen	Chile	-72.4	-46.2	14	15	2015
Araucania	Chile	-72.5	-38.8	14	15	2015
LosRios	Chile	-72.5	-40.1	14	15	2015
LosRios2	Chile	-72.5	-40.1	15	15	2001
LosLagos	Chile	-72.5	-40.1	14	15	2015
AricayPainacota	Chile	-69.6	-18.5	5	5	2001
Auracania	Chile	-72.5	-38.8	15	15	2001
BioBio	Chile	-72.5	-36.8	15	15	2001
SantiagoMetropolitana	Chile	-70.8	-33.3	15	15	2001
SantiagoMetropolitana2	Chile	-70.8	-33.3	21	21	2001
SantiagoMetropolitana3	Chile	-70.8	-33.3	6	7	2017
Valparaiso	Chile	-71.7	-32.9	21	21	2001

Location	Country	Long	Lat	Start Dekad	Final Dekad	Event Year
Valparaiso2	Chile	-71.7	-32.9	6	7	2017
Antofagasta	Chile	-70.5	-23.5	6	7	2017
Coquimbo	Chile	-71.5	-29.8	6	7	2017
Tolima	Colombia	-75.3	4.2	4	6	2000
Santander	Colombia	-73.8	6.8	4	6	2000
Valle	Colombia	-76.8	3.9	4	6	2000
Cordoba	Colombia	-75.7	8.2	31	33	2002
SiloDistrict	Colombia	-72.9	7.3	33	34	2016
Cundinamarca	Colombia	-74.2	5.1	6	7	2017
Huila	Colombia	-75.6	2.7	6	7	2017
Antioquia	Colombia	-75.5	7.3	6	7	2017
Cauca	Colombia	-76.9	2.8	31	31	2017
Salto	Uruguay	-58.1	-31.3	15	15	2000
AllDepartmentsUy	Uruguay	-55.9	-32.4	11	14	2016
Artigas	Uruguay	-56.6	-30.3	16	18	2001
Rivera	Uruguay	-55.7	-30.8	16	18	2001
Durazno	Uruguay	-56.6	-33.3	16	18	2001
Artigas2	Uruguay	-56.6	-30.3	29	30	2015
Salto2	Uruguay	-58.1	-31.3	29	30	2015
Rivera2	Uruguay	-55.7	-30.8	29	30	2015
Tacuarembo	Uruguay	-56.1	-31.6	29	30	2015
Paysandu	Uruguay	-58.2	-32.1	29	30	2015
CerroLargo	Uruguay	-54.3	-32.2	29	30	2015
Canelones	Uruguay	-56.4	-34.4	10	10	2002
Durazno2	Uruguay	-56.6	-33.3	10	10	2002
Florida	Uruguay	-56.3	-34.0	10	10	2002
Montevideo	Uruguay	-56.3	-34.8	10	10	2002
Rocha	Uruguay	-54.5	-34.4	10	10	2002
SanJose	Uruguay	-56.8	-34.2	10	10	2002
Guerrero	Mexico	-99.7	17.6	25	25	2016
Chiapas	Mexico	-93.2	16.9	25	25	2016
Oaxaca	Mexico	-96.8	17.2	16	17	2000
Chalco	Mexico	-99.0	19.4	15	15	2000
Oaxaca2	Mexico	-96.8	17.2	13	13	2005
Guerrero2	Mexico	-99.7	17.6	26	27	2016
Chiapas2	Mexico	-93.2	16.9	26	27	2016
AltaVerapaz	Guatemala	-90.3	15.7	25	30	2017

Location	Country	Long	Lat	Start Dekad	Final Dekad	Event Year
Isabel	Guatemala	-89.3	15.5	22	22	2015
Retalhuleu	Guatemala	-91.8	14.6	26	26	2017
Atlantida	Honduras	-87.3	15.8	1	3	2002
Cortes	Honduras	-88.1	15.6	26	26	2003
Atlantida2	Honduras	-87.3	15.8	31	31	2007
Colon	Honduras	-85.5	15.8	31	31	2007
FranciscoMorazan	Honduras	-87.2	14.6	29	29	2016
Valle	Honduras	-87.7	13.7	29	29	2016
Choluteca	Honduras	-87.3	13.4	29	29	2016
Paraiso	Honduras	-86.5	14.2	29	29	2016
Colon2	Honduras	-85.5	15.8	1	3	2017
AllDepts_Hon	Honduras	-86.4	15.3	30	30	2017
Bocas del Toro	Panama	-82.6	9.5	13	13	2001
Chiriqui	Panama	-82.5	8.7	25	27	2000

6 Annex 2
Alternate Satellite Sources Descriptions

This section presents the details of each of the datasources used for comparison as well as the IRI Data Library links where these datasets can be found.

6.1 CHIRPS

Climate Hazards Group InfraRed Precipitation with Station data (CHIRPS) is a 30+ year quasi-global rainfall dataset. Spanning 50°S–50°N (and all longitudes), starting in 1981 to near-present, CHIRPS incorporates 0.05° resolution satellite imagery with in-situ station data to create gridded rainfall time series for trend analysis and seasonal drought monitoring.

CHIRPS is calculated in a two parts process, first InfraRed Precipitations (IRP) are estimated from satellite data by calculating the percentage of time that IR observations shows cold cloud tops (<235ºK) and converting that percentage into millimeters of precipitation by means of previously determined local regression with TRMM 3B42 precipitation. Then these values are divided by their long term IRP mean (1981-2012). The percent of normal IRP daily is then multiplied by the corresponding Climate Hazards Group's Precipitation Climatology (CHPClim) to produce an unbiased gridded estimate, with units of millimeters per day, called the Climate Hazards Group IR Precipitation (CHIRP). The second part of the process is to add the station observations information to get the final CHIRPS product. To blend CHIRP with the station information, for each grid location of CHIRP, the five nearest stations are used to calculate and adjustment ratio for CHIRP. Each of the five stations is assigned with a weight proportional to square of their expected correlation; afterwards the five weights are scaled to sum 1 and used to blend the station data into a single modifier (ratio) that can be used to adjust the CHIRP estimates.

http://iridl.ldeo.columbia.edu/SOURCES/.UCSB/.CHIRPS/.v2p0/

6.2 CHIRP

Climate Hazards Group InfraRed Precipitation (CHIRP) is a 30+ year quasi-global rainfall decadal dataset. Spanning 50°S–50°N (and all longitudes), starting in 1981 to near-present, CHIRP uses 0.05° resolution satellite imagery to create gridded rainfall time series for trend analysis and seasonal drought monitoring.

To calculate CHIRP, first InfraRed Precipitations (IRP) are estimated from satellite data by calculating the percentage of time that IR observations shows cold cloud tops (<235ºK) and converting that percentage into millimeters of precipitation by means of previously determined local regression with TRMM 3B42 precipitation. Then these values are divided by their long term IRP mean (1981-2012). The percent of normal IRP dekadal is then multiplied by the corresponding CHPClim to produce an unbiased gridded estimate, with units of millimeters per day.

http://iridl.ldeo.columbia.edu/SOURCES/.UCSB/.CHIRP/.v1p0/

6.3 TRMM

Tropical Rainfall Measuring Mission (TRMM) consists of an algorithm, called 3B42, that produces merged high quality (HQ)/infrared (IR) precipitation estimates. These gridded estimates are on a 3-hour temporal resolution and a 0.25-degree by 0.25-degree spatial resolution in a global belt extending from 50 degrees South to 50 degrees North latitude available from March 2000 to May 2015.

The 3B42 estimates are produced in four stages; (1) the microwave precipitation estimates are calibrated and combined, (2) infrared precipitation estimates are created using the calibrated microwave precipitation, (3) the microwave and IR estimates are combined, and (4) rescaling to monthly data is applied.

Note: The Global Precipitation Mission (GPM) is a new satellite precipitation product from NASA; it is a global (65N-65S) gridded rainfall dataset with 0.1 degrees resolution with information every 30 minutes March 2014.

https://iridl.ldeo.columbia.edu/SOURCES/.NASA/.GES-DAAC/.TRMM_L3/.TRMM_3B42/.v7/

6.4 NDVI and EVI

The Enhanced Vegetation Index (EVI) and Normalized Difference Vegetation Index (NDVI) are a vegetation indices that provide a quantitative measure of the amount of photosynthetically active plants on the ground at a particular place and time. Both products have a global coverage with a spatial resolution as fine as 250 m from February 2000 - present, with measurements provided twice a month (16-day temporal resolution).

NDVI is calculated as a ratio of the difference of the Near Infrared (NIR) and the Red bands on a sensor:

$$NDVI=(NIR-Red)/(NIR+Red)$$

EVI uses a similar algorithm but incorporates other terms to be less influence by soil color and shadows, as well as to stabilize the index value against variations in aerosol concentration levels. The following formula is used to calculate EVI:

$$EVI=G*(NIR-Red)/(NIR+C_1*Red-C_2*Blue+L)$$

Where L is the canopy background adjustment for correcting nonlinear, differential NIR and red radiant transfer through a canopy; C1 and C2 are the coefficients of the aerosol resistance term (which uses the blue band to correct for aerosol influences in the red band); and G is a gain or scaling factor. The coefficients adopted in the EVI algorithm are, L=1, C1=6, C2=7.5, and G=2.5.

Because vegetation has high NIR reflectance but low red reflectance, vegetated areas will have higher NDVI and EVI values compared to non-vegetated areas.

http://iridl.ldeo.columbia.edu/SOURCES/.USGS/.LandDAAC/.MODIS/.version_005/

Annex 3
R Script for Dataset Validation

Disclaimer

The script below was developed through a range of projects and different sources, and has been adapted to the needs and specific requirements of this effort. Copyright to this script belongs to Columbia University and is not to be considered one of the deliverables required under this project.

We are sharing the code in order to provide further insight into the types of analysis that were carried out for the validation of the dataset.

R script code description

```
####################################
#  CIF LATAM VALIDATION - DROUGHT  #
####################################

###############
#  Libraries  #
###############
library(dplyr)
library(stargazer) # REQUIRED FOR OUTPUT TABLES
library(pscl) # REQUIRED FOR COMPUTING PSEUDO R2

###############
# Directories #
###############

runname <- "CIF_LATAM" # This name needs to match the runset
in.path<-paste("~","irifist","sniid_runs",runname,"Input", sep="/")
dir.create(paste("~","irifist","sniid_runs",runname,"ValidationResults", sep="/")) # Just run the first time to create the results folder
results.path<-paste("~","irifist","sniid_runs",runname,"ValidationResults", sep="/")

########################
#  Read & Reshape Data  #
########################

#Definition of the years#
BeginYear<-2000
EndYear <- 2016

# SITE TABLE #
originalsitetable<-paste(in.path,"validationsitetable.csv",sep='/') ## Read all the places in the different countries
sitetable<- read.csv(originalsitetable, header=TRUE)
sitetable <- sitetable[,-c(1)]

#############
# BAD YEARS #
#############

BadYears<-matrix(data=0,nrow=nrow(sitetable),ncol=EndYear-BeginYear+3)
colnames(BadYears)<-as.character(c("Name","Region",BeginYear:EndYear))
BadYears<-as.data.frame(BadYears)
BadYears$Name<-as.character(unlist(sitetable$Name))
BadYears$Region<-as.character(unlist(sitetable$Region))

for(sites in 1:length(sitetable$Name)){readbadyears<-sitetable[sites,12:15]
```

```
 # for (j in 1:5){if (readbadyears[j]<BeginYear){readbadyears[j]<-NA}}
 readbadyears<-t(readbadyears); readbadyears<-readbadyears[complete.cases(readbadyears),]; readbadyears<-t(readbadyears)
 for (i in 1:length(readbadyears)) { bdyr<-readbadyears[i];BadYears[sites,as.character(bdyr)]<-1}}

BadYears <- reshape(BadYears, varying= list(names(BadYears)[3:ncol(BadYears)]), v.name="BadYears", idvar="Name",
times=BeginYear:EndYear, timevar = "Year", direction="long")

#################
# LatAm dataset #
#################

# Early #
LatAmEarly<-matrix(data=NA,nrow=nrow(sitetable),ncol=EndYear-BeginYear+3)
colnames(LatAmEarly)<-as.character(c("Name","Region",BeginYear:EndYear))
LatAmEarly<-as.data.frame(LatAmEarly)
LatAmEarly$Name<-as.character(unlist(sitetable$Name))
LatAmEarly$Region<-as.character(unlist(sitetable$Region))

for(sites in 1:length(sitetable$Name)){
 name <- sitetable$Name[sites]
 met.pth <- paste("~","irifist","SNIID_Runs",runname,"scenarios",name,"met.data",sep="/")
 commondata.pth<- paste("~","irifist","sniid_code","Common_Data",sep="/")
 dekas.day.n<-read.csv(paste(commondata.pth,"DekadStartDay.csv", sep="/"), header=TRUE)
 fin <- "LatAm.csv" ; file.name <-paste(met.pth,fin,sep="/")
 rain.in<-read.csv(file.name,header=TRUE,skip=0); rain.in <- rain.in[,-c(1)] # reading in the CHIRP data
 startdek<-sitetable$EarlyFirst[sites] ; enddek<-sitetable$EarlyLast[sites]
 startday<- dekas.day.n[startdek,2] ;endday<- dekas.day.n[enddek,3]
 mean.window.data<-matrix(data=NA,nrow=1,ncol=EndYear-BeginYear+1)
 for (i in 1:ncol(mean.window.data)){ mean.window.data[i]<-mean(rain.in[startday:endday,i],na.rm=TRUE)}
 LatAmEarly[sites,3:ncol(LatAmEarly)]<-mean.window.data}

LatAmEarly <- reshape(LatAmEarly, varying= list(names(LatAmEarly)[3:ncol(LatAmEarly)]), v.name="LatAmEarly", idvar="Name",
times=BeginYear:EndYear, timevar = "Year", direction="long")

# Late #
LatAmLate<-matrix(data=NA,nrow=nrow(sitetable),ncol=EndYear-BeginYear+3)
colnames(LatAmLate)<-as.character(c("Name","Region",BeginYear:EndYear))
LatAmLate<-as.data.frame(LatAmLate)
LatAmLate$Name<-as.character(unlist(sitetable$Name))
LatAmLate$Region<-as.character(unlist(sitetable$Region))

for(sites in 1:length(sitetable$Name)){
 name <- sitetable$Name[sites]
 met.pth <- paste("~","irifist","SNIID_Runs",runname,"scenarios",name,"met.data",sep="/")
 commondata.pth<- paste("~","irifist","sniid_code","Common_Data",sep="/")
 dekas.day.n<-read.csv(paste(commondata.pth,"DekadStartDay.csv", sep="/"), header=TRUE)
 fin <- "LatAm.csv" ; file.name <-paste(met.pth,fin,sep="/")
 rain.in<-read.csv(file.name,header=TRUE,skip=0); rain.in <- rain.in[,-c(1)] # reading in the CHIRP data
 startdek<-sitetable$LateFirst[sites] ; enddek<-sitetable$LateLast[sites]
 startday<- dekas.day.n[startdek,2] ;endday<- dekas.day.n[enddek,3]
 mean.window.data<-matrix(data=NA,nrow=1,ncol=EndYear-BeginYear+1)
 for (i in 1:ncol(mean.window.data)){ mean.window.data[i]<-mean(rain.in[startday:endday,i],na.rm=TRUE)}
 LatAmLate[sites,3:ncol(LatAmLate)]<-mean.window.data}

LatAmLate <- reshape(LatAmLate, varying= list(names(LatAmLate)[3:ncol(LatAmLate)]), v.name="LatAmLate", idvar="Name",
times=BeginYear:EndYear, timevar = "Year", direction="long")

##########
# CHIRPS #
##########

# Early #
CHIRPSEarly<-matrix(data=NA,nrow=nrow(sitetable),ncol=EndYear-BeginYear+3)
colnames(CHIRPSEarly)<-as.character(c("Name","Region",BeginYear:EndYear))
CHIRPSEarly<-as.data.frame(CHIRPSEarly)
CHIRPSEarly$Name<-as.character(unlist(sitetable$Name))
CHIRPSEarly$Region<-as.character(unlist(sitetable$Region))

for(sites in 1:length(sitetable$Name)){
 name <- sitetable$Name[sites]
```

```
 met.pth <- paste("~","irifist","SNIID_Runs",runname,"scenarios",name,"met.data",sep="/")
 commondata.pth<- paste("~","irifist","sniid_code","Common_Data",sep="/")
 dekas.day.n<-read.csv(paste(commondata.pth,"DekadStartDay.csv", sep="/"), header=TRUE)
 fin <- "CHIRPS.csv" ; file.name <-paste(met.pth,fin,sep="/")
 rain.in<-read.csv(file.name,header=TRUE,skip=0); rain.in <- rain.in[,-c(1)] # reading in the CHIRP data
 startdek<-sitetable$EarlyFirst[sites] ; enddek<-sitetable$EarlyLast[sites]
 startday<- dekas.day.n[startdek,2] ;endday<- dekas.day.n[enddek,3]
 mean.window.data<-matrix(data=NA,nrow=1,ncol=EndYear-BeginYear+1)
 for (i in 1:ncol(mean.window.data)){ mean.window.data[i]<-mean(rain.in[startday:endday,i],na.rm=TRUE)}
 CHIRPSEarly[sites,3:ncol(CHIRPSEarly)]<-mean.window.data}

CHIRPSEarly <- reshape(CHIRPSEarly, varying= list(names(CHIRPSEarly)[3:ncol(CHIRPSEarly)]), v.name="CHIRPSEarly", idvar="Name",
times=BeginYear:EndYear, timevar = "Year", direction="long")

# Late #
CHIRPSLate<-matrix(data=NA,nrow=nrow(sitetable),ncol=EndYear-BeginYear+3)
colnames(CHIRPSLate)<-as.character(c("Name","Region",BeginYear:EndYear))
CHIRPSLate<-as.data.frame(CHIRPSLate)
CHIRPSLate$Name<-as.character(unlist(sitetable$Name))
CHIRPSLate$Region<-as.character(unlist(sitetable$Region))

for(sites in 1:length(sitetable$Name)){
 name <- sitetable$Name[sites]
 met.pth <- paste("~","irifist","SNIID_Runs",runname,"scenarios",name,"met.data",sep="/")
 commondata.pth<- paste("~","irifist","sniid_code","Common_Data",sep="/")
 dekas.day.n<-read.csv(paste(commondata.pth,"DekadStartDay.csv", sep="/"), header=TRUE)
 fin <- "CHIRPS.csv" ; file.name <-paste(met.pth,fin,sep="/")
 rain.in<-read.csv(file.name,header=TRUE,skip=0); rain.in <- rain.in[,-c(1)] # reading in the CHIRP data
 startdek<-sitetable$LateFirst[sites] ; enddek<-sitetable$LateLast[sites]
 startday<- dekas.day.n[startdek,2] ;endday<- dekas.day.n[enddek,3]
 mean.window.data<-matrix(data=NA,nrow=1,ncol=EndYear-BeginYear+1)
 for (i in 1:ncol(mean.window.data)){ mean.window.data[i]<-mean(rain.in[startday:endday,i],na.rm=TRUE)}
 CHIRPSLate[sites,3:ncol(CHIRPSLate)]<-mean.window.data}

CHIRPSLate <- reshape(CHIRPSLate, varying= list(names(CHIRPSLate)[3:ncol(CHIRPSLate)]), v.name="CHIRPSLate", idvar="Name",
times=BeginYear:EndYear, timevar = "Year", direction="long")

##########
# CHIRP #
##########

# Early #
CHIRPEarly<-matrix(data=NA,nrow=nrow(sitetable),ncol=EndYear-BeginYear+3)
colnames(CHIRPEarly)<-as.character(c("Name","Region",BeginYear:EndYear))
CHIRPEarly<-as.data.frame(CHIRPEarly)
CHIRPEarly$Name<-as.character(unlist(sitetable$Name))
CHIRPEarly$Region<-as.character(unlist(sitetable$Region))

for(sites in 1:length(sitetable$Name)){
 name <- sitetable$Name[sites]
 met.pth <- paste("~","irifist","SNIID_Runs",runname,"scenarios",name,"met.data",sep="/")
 commondata.pth<- paste("~","irifist","sniid_code","Common_Data",sep="/")
 dekas.day.n<-read.csv(paste(commondata.pth,"DekadStartDay.csv", sep="/"), header=TRUE)
 fin <- "CHIRP.csv" ; file.name <-paste(met.pth,fin,sep="/")
 rain.in<-read.csv(file.name,header=TRUE,skip=0); rain.in <- rain.in[,-c(1)] # reading in the CHIRP data
 startdek<-sitetable$EarlyFirst[sites] ; enddek<-sitetable$EarlyLast[sites]
 startday<- dekas.day.n[startdek,2] ;endday<- dekas.day.n[enddek,3]
 mean.window.data<-matrix(data=NA,nrow=1,ncol=EndYear-BeginYear+1)
 for (i in 1:ncol(mean.window.data)){ mean.window.data[i]<-mean(rain.in[startday:endday,i],na.rm=TRUE)}
 CHIRPEarly[sites,3:ncol(CHIRPEarly)]<-mean.window.data}

CHIRPEarly <- reshape(CHIRPEarly, varying= list(names(CHIRPEarly)[3:ncol(CHIRPEarly)]), v.name="CHIRPEarly", idvar="Name",
times=BeginYear:EndYear, timevar = "Year", direction="long")

# Late #
CHIRPLate<-matrix(data=NA,nrow=nrow(sitetable),ncol=EndYear-BeginYear+3)
colnames(CHIRPLate)<-as.character(c("Name","Region",BeginYear:EndYear))
CHIRPLate<-as.data.frame(CHIRPLate)
CHIRPLate$Name<-as.character(unlist(sitetable$Name))
CHIRPLate$Region<-as.character(unlist(sitetable$Region))
```

```
for(sites in 1:length(sitetable$Name)){
 name <- sitetable$Name[sites]
 met.pth <- paste("~","irifist","SNIID_Runs",runname,"scenarios",name,"met.data",sep="/")
 commondata.pth<- paste("~","irifist","sniid_code","Common_Data",sep="/")
 dekas.day.n<-read.csv(paste(commondata.pth,"DekadStartDay.csv", sep="/"), header=TRUE)
 fin <- "CHIRP.csv" ; file.name <-paste(met.pth,fin,sep="/")
 rain.in<-read.csv(file.name,header=TRUE,skip=0); rain.in <- rain.in[,-c(1)] # reading in the CHIRP data
 startdek<-sitetable$LateFirst[sites] ; enddek<-sitetable$LateLast[sites]
 startday<- dekas.day.n[startdek,2] ;endday<- dekas.day.n[enddek,3]
 mean.window.data<-matrix(data=NA,nrow=1,ncol=EndYear-BeginYear+1)
 for (i in 1:ncol(mean.window.data)){ mean.window.data[i]<-mean(rain.in[startday:endday,i],na.rm=TRUE)}
 CHIRPLate[sites,3:ncol(CHIRPLate)]<-mean.window.data}

CHIRPLate <- reshape(CHIRPLate, varying= list(names(CHIRPLate)[3:ncol(CHIRPLate)]), v.name="CHIRPLate", idvar="Name",
times=BeginYear:EndYear, timevar = "Year", direction="long")

##########
# TRMM #
##########

# Early #
TRMMEarly<-matrix(data=NA,nrow=nrow(sitetable),ncol=EndYear-BeginYear+3)
colnames(TRMMEarly)<-as.character(c("Name","Region",BeginYear:EndYear))
TRMMEarly<-as.data.frame(TRMMEarly)
TRMMEarly$Name<-as.character(unlist(sitetable$Name))
TRMMEarly$Region<-as.character(unlist(sitetable$Region))

for(sites in 1:length(sitetable$Name)){
 name <- sitetable$Name[sites]
 met.pth <- paste("~","irifist","SNIID_Runs",runname,"scenarios",name,"met.data",sep="/")
 commondata.pth<- paste("~","irifist","sniid_code","Common_Data",sep="/")
 dekas.day.n<-read.csv(paste(commondata.pth,"DekadStartDay.csv", sep="/"), header=TRUE)
 fin <- "TRMM.csv" ; file.name <-paste(met.pth,fin,sep="/")
 rain.in<-read.csv(file.name,header=TRUE,skip=0); rain.in <- rain.in[,-c(1)] # reading in the TRMM data
 startdek<-sitetable$EarlyFirst[sites] ; enddek<-sitetable$EarlyLast[sites]
 startday<- dekas.day.n[startdek,2] ;endday<- dekas.day.n[enddek,3]
 mean.window.data<-matrix(data=NA,nrow=1,ncol=EndYear-BeginYear+1)
 for (i in 1:ncol(mean.window.data)){ mean.window.data[i]<-mean(rain.in[startday:endday,i],na.rm=TRUE)}
 TRMMEarly[sites,3:ncol(TRMMEarly)]<-mean.window.data}

TRMMEarly <- reshape(TRMMEarly, varying= list(names(TRMMEarly)[3:ncol(TRMMEarly)]), v.name="TRMMEarly", idvar="Name",
times=BeginYear:EndYear, timevar = "Year", direction="long")

# Late #
TRMMLate<-matrix(data=NA,nrow=nrow(sitetable),ncol=EndYear-BeginYear+3)
colnames(TRMMLate)<-as.character(c("Name","Region",BeginYear:EndYear))
TRMMLate<-as.data.frame(TRMMLate)
TRMMLate$Name<-as.character(unlist(sitetable$Name))
TRMMLate$Region<-as.character(unlist(sitetable$Region))

for(sites in 1:length(sitetable$Name)){
 name <- sitetable$Name[sites]
 met.pth <- paste("~","irifist","SNIID_Runs",runname,"scenarios",name,"met.data",sep="/")
 commondata.pth<- paste("~","irifist","sniid_code","Common_Data",sep="/")
 dekas.day.n<-read.csv(paste(commondata.pth,"DekadStartDay.csv", sep="/"), header=TRUE)
 fin <- "TRMM.csv" ; file.name <-paste(met.pth,fin,sep="/")
 rain.in<-read.csv(file.name,header=TRUE,skip=0); rain.in <- rain.in[,-c(1)] # reading in the TRMM data
 startdek<-sitetable$LateFirst[sites] ; enddek<-sitetable$LateLast[sites]
 startday<- dekas.day.n[startdek,2] ;endday<- dekas.day.n[enddek,3]
 mean.window.data<-matrix(data=NA,nrow=1,ncol=EndYear-BeginYear+1)
 for (i in 1:ncol(mean.window.data)){ mean.window.data[i]<-mean(rain.in[startday:endday,i],na.rm=TRUE)}
 TRMMLate[sites,3:ncol(TRMMLate)]<-mean.window.data}

TRMMLate <- reshape(TRMMLate, varying= list(names(TRMMLate)[3:ncol(TRMMLate)]), v.name="TRMMLate", idvar="Name",
times=BeginYear:EndYear, timevar = "Year", direction="long")

##########
# NDVI #
##########
```

```
# Early #
NDVIEarly<-matrix(data=NA,nrow=nrow(sitetable),ncol=EndYear-BeginYear+3)
colnames(NDVIEarly)<-as.character(c("Name","Region",BeginYear:EndYear))
NDVIEarly<-as.data.frame(NDVIEarly)
NDVIEarly$Name<-as.character(unlist(sitetable$Name))
NDVIEarly$Region<-as.character(unlist(sitetable$Region))

for(sites in 1:length(sitetable$Name)){
 name <- sitetable$Name[sites]
 met.pth <- paste("~","irifist","SNIID_Runs",runname,"scenarios",name,"met.data",sep="/")
 fin <- "NDVI.csv" ; file.name <-paste(met.pth,fin,sep="/")
 rain.in<-read.csv(file.name,header=TRUE,skip=0); rain.in <- rain.in[,-c(1)] # reading in the NDVI data
 startdek<-sitetable$EarlyFirst[sites] ; enddek<-sitetable$EarlyLast[sites]
 mean.window.data<-matrix(data=NA,nrow=1,ncol=EndYear-BeginYear+1)
 for (i in 1:ncol(mean.window.data)){ mean.window.data[i]<-mean(rain.in[startdek:enddek,i],na.rm=TRUE)}
 NDVIEarly[sites,3:ncol(NDVIEarly)]<-mean.window.data}

NDVIEarly <- reshape(NDVIEarly, varying= list(names(NDVIEarly)[3:ncol(NDVIEarly)]), v.name="NDVIEarly", idvar="Name",
times=BeginYear:EndYear, timevar = "Year", direction="long")

# Late #
NDVILate<-matrix(data=NA,nrow=nrow(sitetable),ncol=EndYear-BeginYear+3)
colnames(NDVILate)<-as.character(c("Name","Region",BeginYear:EndYear))
NDVILate<-as.data.frame(NDVILate)
NDVILate$Name<-as.character(unlist(sitetable$Name))
NDVILate$Region<-as.character(unlist(sitetable$Region))

for(sites in 1:length(sitetable$Name)){
 name <- sitetable$Name[sites]
 met.pth <- paste("~","irifist","SNIID_Runs",runname,"scenarios",name,"met.data",sep="/")
 fin <- "NDVI.csv" ; file.name <-paste(met.pth,fin,sep="/")
 rain.in<-read.csv(file.name,header=TRUE,skip=0); rain.in <- rain.in[,-c(1)] # reading in the NDVI data
 startdek<-sitetable$LateFirst[sites] ; enddek<-sitetable$LateLast[sites]
 mean.window.data<-matrix(data=NA,nrow=1,ncol=EndYear-BeginYear+1)
 for (i in 1:ncol(mean.window.data)){ mean.window.data[i]<-mean(rain.in[startdek:enddek,i],na.rm=TRUE)}
 NDVILate[sites,3:ncol(NDVILate)]<-mean.window.data}

NDVILate <- reshape(NDVILate, varying= list(names(NDVILate)[3:ncol(NDVILate)]), v.name="NDVILate", idvar="Name",
times=BeginYear:EndYear, timevar = "Year", direction="long")

##########
#  EVI   #
##########

# Early #
EVIEarly<-matrix(data=NA,nrow=nrow(sitetable),ncol=EndYear-BeginYear+3)
colnames(EVIEarly)<-as.character(c("Name","Region",BeginYear:EndYear))
EVIEarly<-as.data.frame(EVIEarly)
EVIEarly$Name<-as.character(unlist(sitetable$Name))
EVIEarly$Region<-as.character(unlist(sitetable$Region))

for(sites in 1:length(sitetable$Name)){
 name <- sitetable$Name[sites]
 met.pth <- paste("~","irifist","SNIID_Runs",runname,"scenarios",name,"met.data",sep="/")
 fin <- "EVI.csv" ; file.name <-paste(met.pth,fin,sep="/")
 rain.in<-read.csv(file.name,header=TRUE,skip=0); rain.in <- rain.in[,-c(1)] # reading in the EVI data
 startdek<-sitetable$EarlyFirst[sites] ; enddek<-sitetable$EarlyLast[sites]
 mean.window.data<-matrix(data=NA,nrow=1,ncol=EndYear-BeginYear+1)
 for (i in 1:ncol(mean.window.data)){ mean.window.data[i]<-mean(rain.in[startdek:enddek,i],na.rm=TRUE)}
 EVIEarly[sites,3:ncol(EVIEarly)]<-mean.window.data}

EVIEarly <- reshape(EVIEarly, varying= list(names(EVIEarly)[3:ncol(EVIEarly)]), v.name="EVIEarly", idvar="Name",
times=BeginYear:EndYear, timevar = "Year", direction="long")

# Late #
EVILate<-matrix(data=NA,nrow=nrow(sitetable),ncol=EndYear-BeginYear+3)
colnames(EVILate)<-as.character(c("Name","Region",BeginYear:EndYear))
EVILate<-as.data.frame(EVILate)
EVILate$Name<-as.character(unlist(sitetable$Name))
```

```
EVILate$Region<-as.character(unlist(sitetable$Region))

for(sites in 1:length(sitetable$Name)){
 name <- sitetable$Name[sites]
 met.pth <- paste("~","irifist","SNIID_Runs",runname,"scenarios",name,"met.data",sep="/")
 fin <- "EVI.csv" ; file.name <-paste(met.pth,fin,sep="/")
 rain.in<-read.csv(file.name,header=TRUE,skip=0); rain.in <- rain.in[,-c(1)] # reading in the EVI data
 startdek<-sitetable$LateFirst[sites] ; enddek<-sitetable$LateLast[sites]
 mean.window.data<-matrix(data=NA,nrow=1,ncol=EndYear-BeginYear+1)
 for (i in 1:ncol(mean.window.data)){ mean.window.data[i]<-mean(rain.in[startdek:enddek,i],na.rm=TRUE)}
 EVILate[sites,3:ncol(EVILate)]<-mean.window.data}

EVILate <- reshape(EVILate, varying= list(names(EVILate)[3:ncol(EVILate)]), v.name="EVILate", idvar="Name", times=BeginYear:EndYear,
timevar = "Year", direction="long")

####################
#  Merge Datasets  #
####################
# Merge
validation_panel <- merge(BadYears, LatAmEarly, by=c("Name", "Region", "Year"))
validation_panel <- merge(validation_panel, LatAmLate, by=c("Name", "Region", "Year"))
validation_panel <- merge(validation_panel, CHIRPSEarly, by=c("Name", "Region", "Year"))
validation_panel <- merge(validation_panel, CHIRPSLate, by=c("Name", "Region", "Year"))
validation_panel <- merge(validation_panel, CHIRPEarly, by=c("Name", "Region", "Year"))
validation_panel <- merge(validation_panel, CHIRPLate, by=c("Name", "Region", "Year"))
validation_panel <- merge(validation_panel, TRMMEarly, by=c("Name", "Region", "Year"))
validation_panel <- merge(validation_panel, TRMMLate, by=c("Name", "Region", "Year"))
validation_panel <- merge(validation_panel, NDVIEarly, by=c("Name", "Region", "Year"))
validation_panel <- merge(validation_panel, NDVILate, by=c("Name", "Region", "Year"))
validation_panel <- merge(validation_panel, EVIEarly, by=c("Name", "Region", "Year"))
validation_panel <- merge(validation_panel, EVILate, by=c("Name", "Region", "Year"))

####################
# Binary Matching  #
####################

# Ranking based on datasets
validation_panel <- validation_panel %>% group_by(Name) %>% mutate(LatAmEarlyRank = order(order(LatAmEarly, decreasing=FALSE)))
validation_panel <- validation_panel %>% group_by(Name) %>% mutate(LatAmLateRank = order(order(LatAmLate, decreasing=FALSE)))
validation_panel <- validation_panel %>% group_by(Name) %>% mutate(CHIRPSEarlyRank = order(order(CHIRPSEarly,
decreasing=FALSE)))
validation_panel <- validation_panel %>% group_by(Name) %>% mutate(CHIRPSLateRank = order(order(CHIRPSLate, decreasing=FALSE)))
validation_panel <- validation_panel %>% group_by(Name) %>% mutate(CHIRPEarlyRank = order(order(CHIRPEarly, decreasing=FALSE)))
validation_panel <- validation_panel %>% group_by(Name) %>% mutate(CHIRPLateRank = order(order(CHIRPLate, decreasing=FALSE)))
validation_panel <- validation_panel %>% group_by(Name) %>% mutate(TRMMEarlyRank = order(order(TRMMEarly, decreasing=FALSE)))
validation_panel <- validation_panel %>% group_by(Name) %>% mutate(TRMMLateRank = order(order(TRMMLate, decreasing=FALSE)))
validation_panel <- validation_panel %>% group_by(Name) %>% mutate(NDVIEarlyRank = order(order(NDVIEarly, decreasing=FALSE)))
validation_panel <- validation_panel %>% group_by(Name) %>% mutate(NDVILateRank = order(order(NDVILate, decreasing=FALSE)))
validation_panel <- validation_panel %>% group_by(Name) %>% mutate(EVIEarlyRank = order(order(EVIEarly, decreasing=FALSE)))
validation_panel <- validation_panel %>% group_by(Name) %>% mutate(EVILateRank = order(order(EVILate, decreasing=FALSE)))

# Macthing Bad Years with top X ranked Years from satelite products
n_rank_yrs<-5

validation_panel$LatAmEarlyMatch <-ifelse(validation_panel$LatAmEarlyRank<=n_rank_yrs & validation_panel$BadYears==1,1,0)
validation_panel$LatAmLateMatch <-ifelse(validation_panel$LatAmLateRank<=n_rank_yrs & validation_panel$BadYears==1,1,0)
validation_panel$CHIRPSEarlyMatch <-ifelse(validation_panel$CHIRPSEarlyRank<=n_rank_yrs & validation_panel$BadYears==1,1,0)
validation_panel$CHIRPSLateMatch <-ifelse(validation_panel$CHIRPSLateRank<=n_rank_yrs & validation_panel$BadYears==1,1,0)
validation_panel$CHIRPEarlyMatch <-ifelse(validation_panel$CHIRPEarlyRank<=n_rank_yrs & validation_panel$BadYears==1,1,0)
validation_panel$CHIRPLateMatch <-ifelse(validation_panel$CHIRPLateRank<=n_rank_yrs & validation_panel$BadYears==1,1,0)
validation_panel$TRMMEarlyMatch <-ifelse(validation_panel$TRMMEarlyRank<=n_rank_yrs & validation_panel$BadYears==1,1,0)
validation_panel$TRMMLateMatch <-ifelse(validation_panel$TRMMLateRank<=n_rank_yrs & validation_panel$BadYears==1,1,0)
validation_panel$NDVIEarlyMatch <-ifelse(validation_panel$NDVIEarlyRank<=n_rank_yrs & validation_panel$BadYears==1,1,0)
validation_panel$NDVILateMatch <-ifelse(validation_panel$NDVILateRank<=n_rank_yrs & validation_panel$BadYears==1,1,0)
validation_panel$EVIEarlyMatch <-ifelse(validation_panel$EVIEarlyRank<=n_rank_yrs & validation_panel$BadYears==1,1,0)
validation_panel$EVILateMatch <-ifelse(validation_panel$EVILateRank<=n_rank_yrs & validation_panel$BadYears==1,1,0)

# Export Results
write.csv(BinaryMatchingCountry, file= paste(results.path,"BinaryMatchingCountry.csv", sep="/"))
write.csv(BinaryMatchingVillage, paste(results.path,"BinaryMatchingVillage.csv", sep="/"))
```

```
write.csv(BinaryMatchingCountryPerc,  paste(results.path,"BinaryMatchingCountryPerc.csv", sep="/"))
write.csv(BinaryMatchingVillagePerc, paste(results.path,"BinaryMatchingVillagePerc.csv", sep="/"))
write.csv(validation_panel, paste(results.path,"validation_panel.csv", sep="/"))

write.csv(BinaryMatchingCountryBoth, file= paste(results.path,"BinaryMatchingCountryBoth.csv", sep="/"))
write.csv(BinaryMatchingVillageBoth, paste(results.path,"BinaryMatchingVillageBoth.csv", sep="/"))
write.csv(BinaryMatchingCountryBothPerc,  paste(results.path,"BinaryMatchingCountryBothPerc.csv", sep="/"))

####################
#     Logits     #
####################
## The R2 command in each country gives a vector of length 6 containing: llh= The log-likelihood from the fitted model; llhNull= The log-likelihood from the intercept-only restricted model; G2=Minus two times the difference in the log-likelihoods; McFadden= McFadden's pseudo r-squared; r2ML=Maximum likelihood pseudo r-squared; r2CU=Cragg and Uhler's pseudo r-squared
## stratgazer is used to export tables in a "paper" format; you can choos .htm or .txt.

unique_countries<- unique(validation_panel$Region)

for (countries in 1:length(unique_countries)) {

logitLatAm<-glm(BadYears ~ LatAmEarly + LatAmLate, data=validation_panel, subset=(Region==unique_countries[countries]),
family="binomial", na.action = na.omit )
R2LatAm<-pR2(logitLatAm)

logitCHIRPS<-glm(BadYears ~ CHIRPSEarly + CHIRPSLate, data=validation_panel, subset=(Region==unique_countries[countries]),
family="binomial", na.action = na.omit )
R2CHIRPS<-pR2(logitCHIRPS)

logitCHIRP<-glm(BadYears ~ CHIRPEarly + CHIRPLate, data=validation_panel, subset=(Region==unique_countries[countries]),
family="binomial", na.action = na.omit )
R2CHIRP<-pR2(logitCHIRP)

logitTRMM<-glm(BadYears ~ TRMMEarly + TRMMLate, data=validation_panel, subset=(Region==unique_countries[countries]),
family="binomial", na.action = na.omit )
R2TRMM<-pR2(logitTRMM)

logitNDVI<-glm(BadYears ~ NDVIEarly + NDVILate, data=validation_panel, subset=(Region==unique_countries[countries]),
family="binomial", na.action = na.omit )
R2NDVI<-pR2(logitNDVI)

logitEVI<-glm(BadYears ~ EVIEarly + EVILate, data=validation_panel, subset=(Region==unique_countries[countries]), family="binomial",
na.action = na.omit )
R2EVI<-pR2(logitEVI)

stargazer(logitLatAm, logitCHIRPS, logitCHIRP, logitTRMM, logitNDVI, logitEVI, type="text", dep.var.labels=c("Bad Years"),
out=paste(results.path,paste("logit",unique_countries[countries],".htm",sep=""), sep="/")) }
```

What is the code doing?

The code has two main features:

The first one is to rank the years according to its precipitation value (in two different season each year), from dry to wet[12], to compare with the bad years gathered from the field. The output is a table of the total hits and misses. In the code it is possible to add multiple datasets and perform this analysis in each of them. The method can be carried out using different time scales (e.g. annual, seasonal or by coverage window), and spatial scales (with different levels of spatial aggregation within the different regions of a country). We analyze two different spatial scales, approximately 625 Km^2 (equivalent to 1 pixel of the LatAm dataset) and 5,625 Km^2 (equivalent to 3 by 3 pixels of the LatAm dataset).

[12] For excess rainfall the code is exactly the same but instead of ranking from dry to wet, it does it from wet to dry.

Secondly, the code runs a logit regression on the datasets, which seek to estimate the change in the probability of a binary variable passing from 0 to 1, by disturbing a continuous variable. In this case, the perception of bad years from the field as a dependent variable is used, and the different dataset studied as an explanatory variable. In this way, the results presented are intended to identify what percentage of the variation in the perception of bad years is explained by the datasets.

i. **What are bad years?**

Bad years are years that have been mentioned by producers as key years were a specific risk has affected their production. In other words, bad years are a set of years where the farmers would have needed to receive a payout from an insurance product.

ii. **What are the other datasets we are using to benchmark against? Why?**

CHIRPS, CHIRP, TRMM, NDVI and EVI.

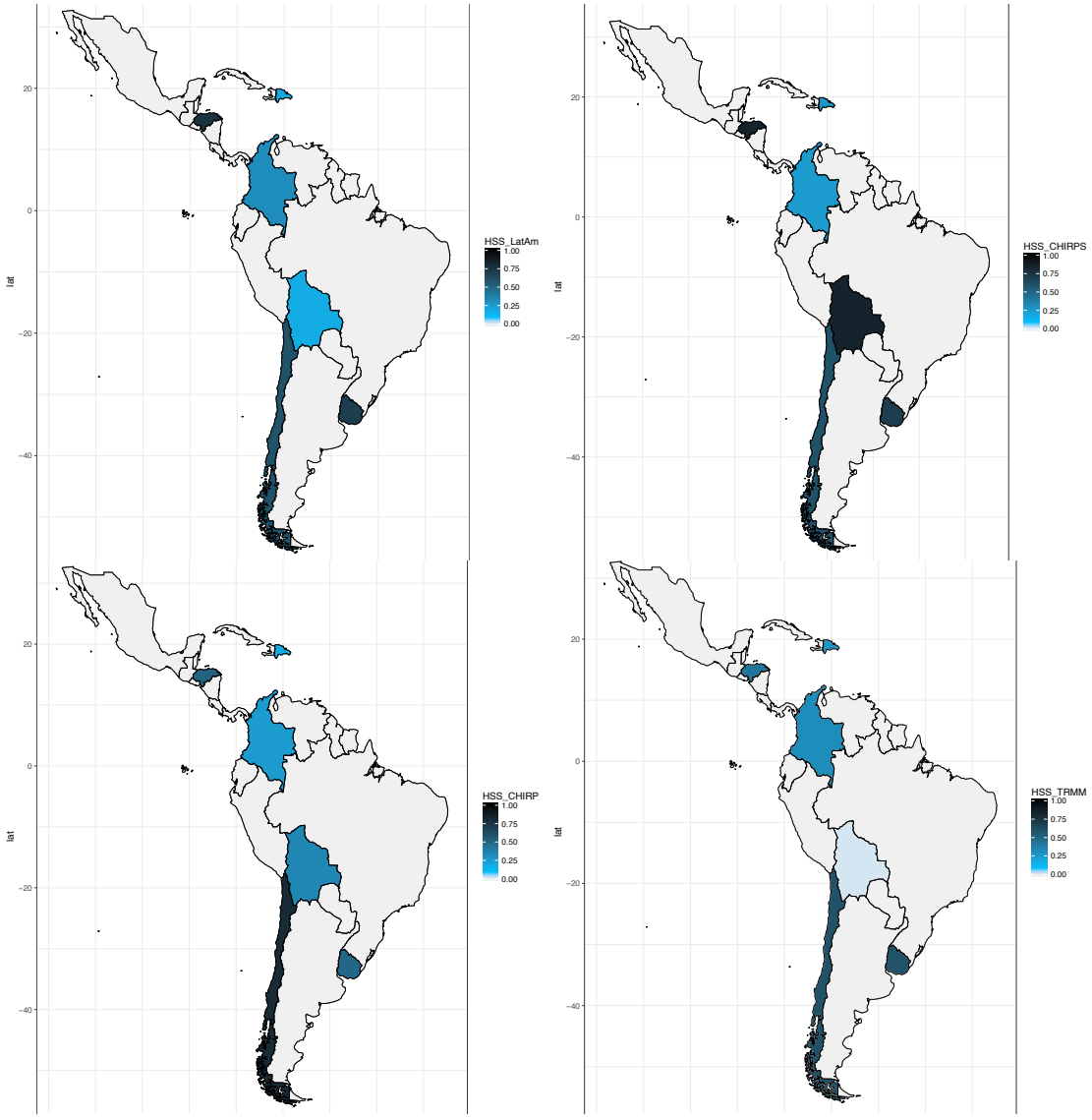
The objective of evaluating other dataset is to evaluate whether the LatAm Dataset is performing better or worse than other datasets available when it comes to capturing bad years, and therefore be a better dataset to base index insurance schemes.

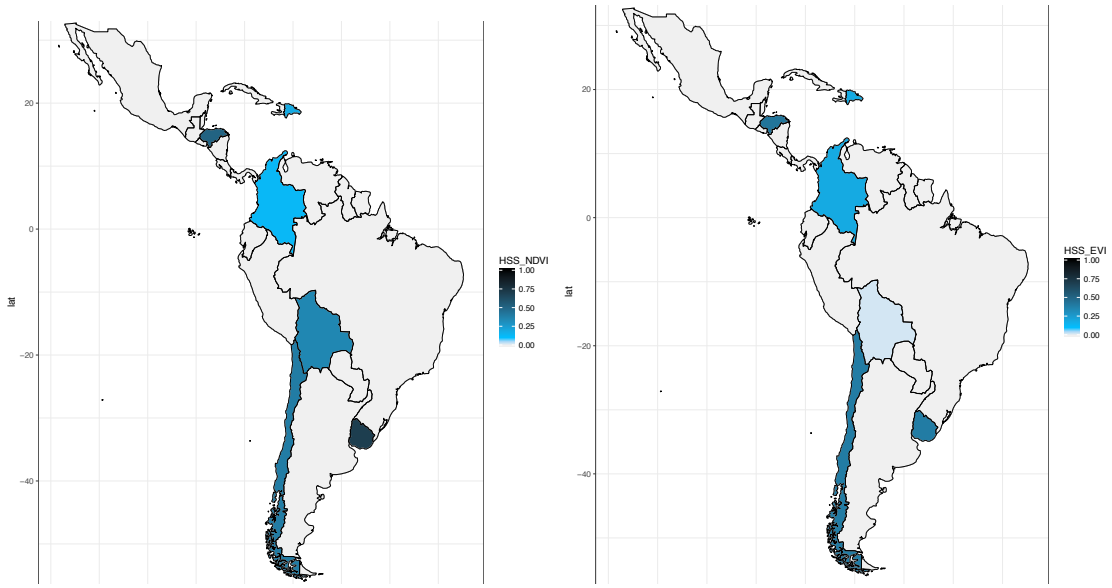
Which risks will it focus on?

The current state of the script focuses on testing the dataset validity for recognizing relevant drought events. The final validation test will focus on both drought and excess rainfall events (for prolonged periods of time—not considering floods.)

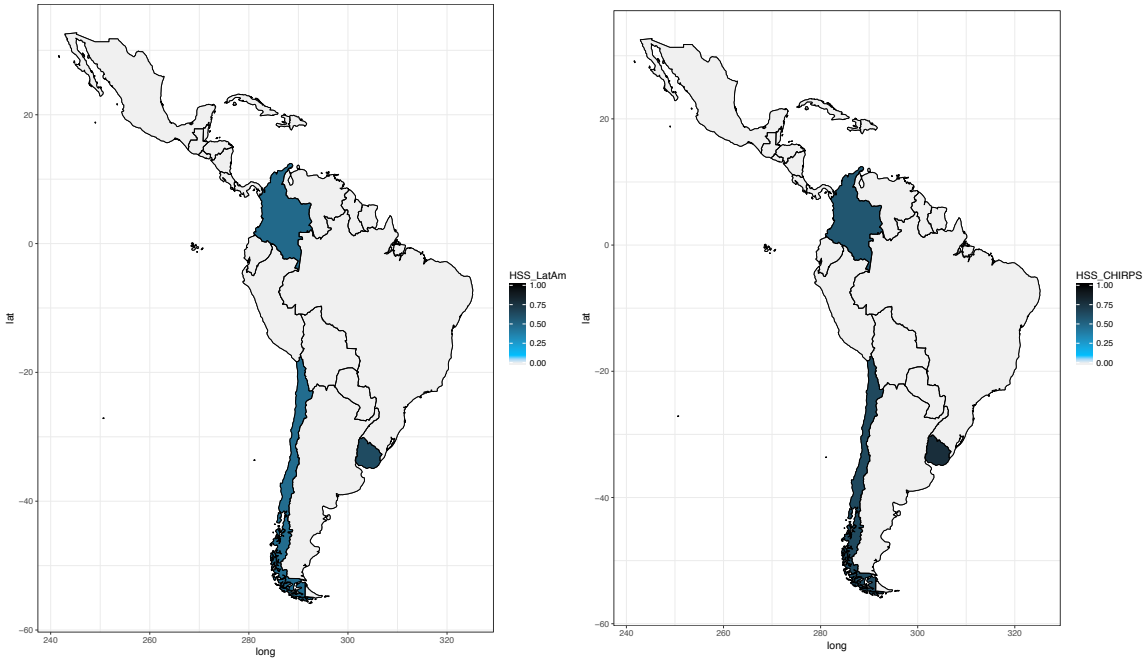
Annex 4
Heidke Skill Scores maps

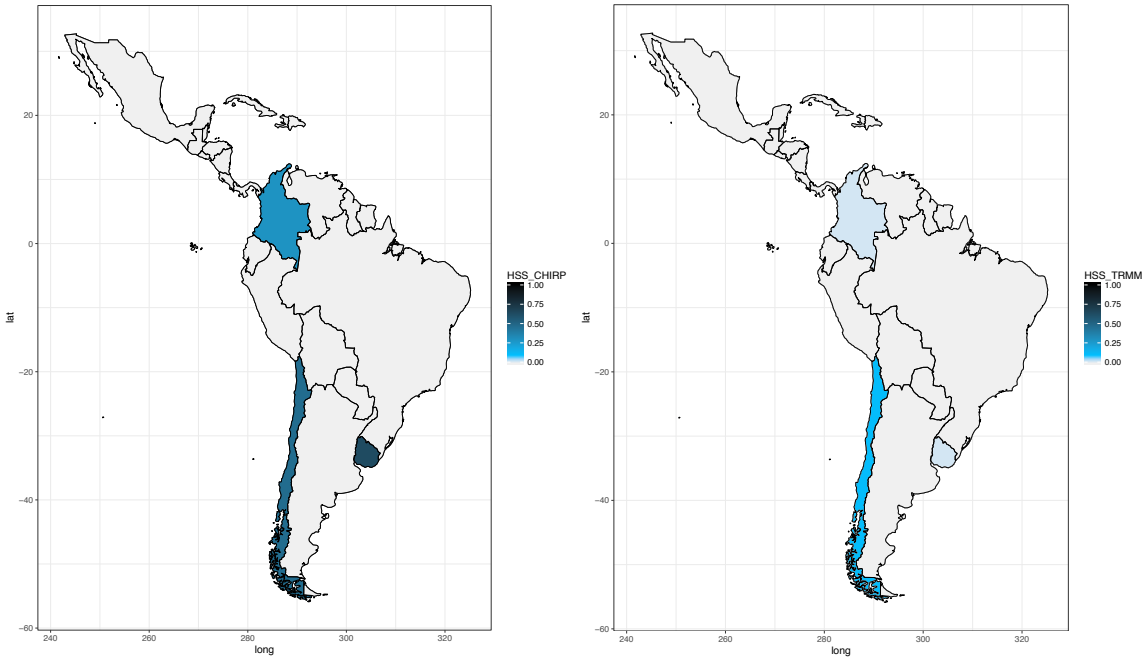
The Heidke Skill Scores for each dataset for the drought risk are displayed in the below maps. The first row contains the map corresponding to the LatAm and CHIRPS datasets, the middle row contains the map corresponding to the CHIRP and TRMM datasets and the bottom row contains the map corresponding to the NDVI and EVI datasets.





The Heidke Skill Scores for each dataset for the excess rainfall risk are displayed in the below maps. The first row contains the map corresponding to the LatAm and CHIRPS datasets and the bottom row contains the map corresponding to the CHIRP and TRMM.




HSS_CHIRP
1.00
0.75
0.50
0.25
0.00
lat
long
20
0
−20
−40
−60
240
260
280
300
320
HSS_TRMM
1.00
0.75
0.50
0.25
0.00
lat
long

Annex 5
Logistic Regression Tables for each Dataset and risk

Table 19: Drought Logit regression Colombia – 625 and 5,625 Km^2 resolution respectively

	Dependent variable:					
	Bad Years					
	(1)	(2)	(3)	(4)	(5)	(6)
LatAmEarly	0.072					
	(0.130)					
LatAmLate	-0.305**					
	(0.143)					
CHIRPSEarly		0.010				
		(0.012)				
CHIRPSLate		-0.019				
		(0.016)				
CHIRPEarly			0.014			
			(0.016)			
CHIRPLate			-0.057*			
			(0.031)			
TRMMEarly				0.106		
				(0.122)		
TRMMLate				-0.259		
				(0.174)		
NDVIEarly					2.405	
					(3.290)	
NDVILate					2.616	
					(3.500)	
EVIEarly						8.000
						(4.883)
EVILate						-0.599
						(4.500)
Constant	-0.210	-0.665	1.419	-0.442	-4.272*	-4.029*
	(0.838)	(1.115)	(1.893)	(1.327)	(2.183)	(2.293)
Observations	102	102	102	90	102	102
Log Likelihood	-47.716	-49.239	-48.172	-40.421	-49.406	-48.959
Akaike Inf. Crit.	101.431	104.478	102.344	86.841	104.813	103.917
Note:						*p<0.1; **p<0.05; ***p<0.01

	Dependent variable:					
	Bad Years					
	(1)	(2)	(3)	(4)	(5)	(6)
LatAmEarly	0.092					
	(0.146)					
LatAmLate	-0.310*					
	(0.159)					
CHIRPSEarly		0.014				
		(0.012)				
CHIRPSLate		-0.023				
		(0.018)				
CHIRPEarly			0.016			
			(0.015)			
CHIRPLate			-0.053			
			(0.034)			
TRMMEarly				0.119		
				(0.118)		
TRMMLate				-0.270		
				(0.175)		
NDVIEarly					6.025	
					(4.566)	
NDVILate					0.288	
					(3.372)	
EVIEarly						16.046**
						(7.587)
EVILate						-1.315
						(4.803)
Constant	-0.193	-0.610	1.217	-0.367	-4.833*	-6.697**
	(1.073)	(1.246)	(2.192)	(1.389)	(2.623)	(3.213)
Observations	102	102	102	90	102	102
Log Likelihood	-48.024	-48.927	-48.668	-40.274	-49.348	-47.833
Akaike Inf. Crit.	102.047	103.855	103.337	86.548	104.696	101.666
Note:						*p<0.1; **p<0.05; ***p<0.01

Table 20: Excess rain Logit regression Colombia – 625 and 5,625 Km^2 resolution respectively

	Dependent variable:			
	Bad Years			
	(1)	(2)	(3)	(4)
LatAmEarly	0.058			
	(0.123)			
LatAmLate	0.065			
	(0.108)			
CHIRPSEarly		0.025*		
		(0.014)		
CHIRPSLate		0.012		
		(0.014)		
CHIRPEarly			0.010	
			(0.018)	
CHIRPLate			-0.003	
			(0.018)	
TRMMEarly				0.192
				(0.148)
TRMMLate				0.034
				(0.141)
Constant	-1.715**	-3.417***	-1.290	-2.337**
	(0.691)	(1.070)	(1.239)	(1.043)
Observations	108	108	108	90
Log Likelihood	-62.228	-59.529	-62.674	-55.044
Akaike Inf. Crit.	130.456	125.058	131.349	116.088
Note:		*p<0.1; **p<0.05; ***p<0.01		

	Dependent variable:			
	Bad Years			
	(1)	(2)	(3)	(4)
LatAmEarly	0.115			
	(0.141)			
LatAmLate	0.072			
	(0.133)			
CHIRPSEarly		0.021		
		(0.013)		
CHIRPSLate		0.005		
		(0.014)		
CHIRPEarly			0.012	
			(0.018)	
CHIRPLate			-0.007	
			(0.021)	
TRMMEarly				0.195
				(0.147)
TRMMLate				-0.004
				(0.147)
Constant	-2.097**	-2.759***	-1.194	-2.161**
	(0.894)	(0.969)	(1.256)	(1.083)
Observations	108	108	108	90
Log Likelihood	-62.003	-60.488	-62.597	-55.251
Akaike Inf. Crit.	130.006	126.975	131.195	116.502
Note:		*p<0.1; **p<0.05; ***p<0.01		

Table 21: Drought Logit regression Dominican Republic – 625 and 5,625 Km^2 resolution respectively

	Dependent variable:					
	Bad Years					
	(1)	(2)	(3)	(4)	(5)	(6)
LatAmEarly	0.163					
	(0.220)					
LatAmLate	-0.121					
	(0.177)					
CHIRPSEarly		-0.018				
		(0.038)				
CHIRPSLate		-0.001				
		(0.020)				
CHIRPEarly			0.076			
			(0.064)			
CHIRPLate			-0.039			
			(0.030)			
TRMMEarly				-0.112		
				(0.282)		
TRMMLate				-0.162		
				(0.223)		
NDVIEarly					-2.877	
					(5.793)	
NDVILate					2.932	
					(4.565)	
EVIEarly						-0.565
						(8.270)
EVILate						2.710
						(5.418)
Constant	-1.044	-0.902	-1.411	0.029	-1.566	-2.294
	(1.015)	(0.883)	(0.929)	(1.331)	(3.498)	(3.406)
Observations	68	68	68	60	68	68
Log Likelihood	-34.172	-34.400	-33.606	-32.142	-34.328	-34.448
Akaike Inf. Crit.	74.343	74.799	73.212	70.285	74.656	74.895
Note:						*p<0.1; **p<0.05; ***p<0.01

	Dependent variable:					
	Bad Years					
	(1)	(2)	(3)	(4)	(5)	(6)
LatAmEarly	0.274					
	(0.256)					
LatAmLate	-0.141					
	(0.234)					
CHIRPSEarly		-0.040				
		(0.043)				
CHIRPSLate		-0.008				
		(0.029)				
CHIRPEarly			0.068			
			(0.077)			
CHIRPLate			-0.037			
			(0.036)			
TRMMEarly				-0.045		
				(0.283)		
TRMMLate				-0.143		
				(0.241)		
NDVIEarly					-9.673	
					(8.919)	
NDVILate					18.088*	
					(10.033)	
EVIEarly						-4.141
						(12.247)
EVILate						24.606*
						(12.860)
Constant	-1.199	-0.093	-1.204	-0.269	-7.622	-10.488*
	(1.381)	(1.684)	(1.603)	(1.437)	(6.098)	(6.283)
Observations	68	68	68	60	68	68
Log Likelihood	-33.912	-34.103	-33.928	-32.382	-32.718	-32.480
Akaike Inf. Crit.	73.824	74.205	73.856	70.765	71.437	70.960
Note:						*p<0.1; **p<0.05; ***p<0.01

Table 22: Drought Logit regression Dominican Republic – Original payout from the NDVI index insurance

VARIABLE DEPENDIENTE: AÑOS MALOS				
VENTANA	Ene - Abr	Jul - Oct	Total	Ambos
Pago	0.0248**	0.0167		0.0214***
	(0.00973)	(0.0112)		(0.00722)
Pago Total			0.0192**	
			(0.00838)	
Constante	-2.365***	-2.428***	-2.205***	-2.399***
	(0.513)	(0.524)	(0.523)	(0.367)
Observaciones	56	56	56	112
Años malos mencionado	8	6	10	14
Años malos capturados por el índice	5 [62.5%]	2 [33.3%]	7 [70.0%]	7 [50.0%]

Errores estándar en paréntesis; *** p<0.01, ** p<0.05, * p<0.1; Porcentaje de años capturados en corchetes.

Table 23: Drought Logit regression Honduras – 625 and 5,625 Km^2 resolution respectively

	Dependent variable:					
	Bad Years					
	(1)	(2)	(3)	(4)	(5)	(6)
LatAmEarly	-0.443***					
	(0.166)					
LatAmLate	-0.159					
	(0.175)					
CHIRPSEarly		-0.077***				
		(0.024)				
CHIRPSLate		-0.083***				
		(0.031)				
CHIRPEarly			-0.050**			
			(0.022)			
CHIRPLate			-0.007			
			(0.018)			
TRMMEarly				-0.285*		
				(0.158)		
TRMMLate				-0.167		
				(0.135)		
NDVIEarly					-18.900***	
					(6.828)	
NDVILate					-2.408	
					(10.081)	
EVIEarly						-12.979
						(10.847)
EVILate						-37.213**
						(15.700)
Constant	1.526	5.624***	2.048	1.706	9.854	18.098***
	(1.042)	(1.788)	(1.599)	(1.291)	(6.203)	(5.778)
Observations	68	68	68	60	68	68
Log Likelihood	-32.512	-22.789	-34.059	-32.389	-31.007	-28.769
Akaike Inf. Crit.	71.024	51.578	74.117	70.777	68.014	63.538
Note:					*p<0.1; **p<0.05; ***p<0.01	

	Dependent variable:					
	Bad Years					
	(1)	(2)	(3)	(4)	(5)	(6)
LatAmEarly	-0.399**					
	(0.164)					
LatAmLate	-0.151					
	(0.174)					
CHIRPSEarly		-0.082***				
		(0.025)				
CHIRPSLate		-0.081***				
		(0.031)				
CHIRPEarly			-0.036*			
			(0.019)			
CHIRPLate			-0.004			
			(0.017)			
TRMMEarly				-0.241		
				(0.157)		
TRMMLate				-0.166		
				(0.139)		
NDVIEarly					-17.721**	
					(7.946)	
NDVILate					-5.826	
					(11.038)	
EVIEarly						-9.945
						(11.065)
EVILate						-45.488***
						(17.177)
Constant	1.366	6.537***	1.264	1.528	12.113*	21.409***
	(1.063)	(2.059)	(1.562)	(1.375)	(7.231)	(6.829)
Observations	68	68	68	60	68	68
Log Likelihood	-33.357	-23.085	-35.089	-32.996	-32.998	-30.124
Akaike Inf. Crit.	72.714	52.170	76.178	71.992	71.995	66.249
Note:					*p<0.1; **p<0.05; ***p<0.01	

Table 24: Drought Logit regression Uruguay – 625 and 5,625 Km² resolution respectively

	Dependent variable:					
	Bad Years					
	(1)	(2)	(3)	(4)	(5)	(6)
LatAmEarly	-0.545*					
	(0.314)					
LatAmLate	-0.852**					
	(0.371)					
CHIRPSEarly		-0.039				
		(0.031)				
CHIRPSLate		-0.072*				
		(0.043)				
CHIRPEarly			-0.065*			
			(0.038)			
CHIRPLate			-0.043			
			(0.042)			
TRMMEarly				-0.181		
				(0.272)		
TRMMLate				-0.664*		
				(0.387)		
NDVIEarly					-33.618***	
					(12.219)	
NDVILate					3.883	
					(11.408)	
EVIEarly						-34.149**
						(14.494)
EVILate						4.356
						(10.257)
Constant	2.881**	2.100*	1.905	1.070	16.768**	9.555*
	(1.321)	(1.270)	(1.476)	(1.228)	(7.360)	(5.147)
Observations	68	68	68	60	68	68
Log Likelihood	-24.253	-26.708	-27.913	-20.426	-26.204	-28.108
Akaike Inf. Crit.	54.506	59.416	61.826	46.853	58.409	62.216
Note:						*p<0.1; **p<0.05; ***p<0.01

	Dependent variable:					
	Bad Years					
	(1)	(2)	(3)	(4)	(5)	(6)
LatAmEarly	-0.588*					
	(0.335)					
LatAmLate	-0.888**					
	(0.398)					
CHIRPSEarly		-0.043				
		(0.032)				
CHIRPSLate		-0.068				
		(0.043)				
CHIRPEarly			-0.068*			
			(0.038)			
CHIRPLate			-0.043			
			(0.042)			
TRMMEarly				-0.162		
				(0.266)		
TRMMLate				-0.700*		
				(0.392)		
NDVIEarly					-41.858***	
					(13.317)	
NDVILate					-4.729	
					(13.357)	
EVIEarly						-38.821***
						(14.514)
EVILate						3.183
						(11.046)
Constant	3.130**	2.122*	1.995	1.142	27.633***	12.068**
	(1.388)	(1.264)	(1.479)	(1.251)	(10.232)	(5.719)
Observations	68	68	68	60	68	68
Log Likelihood	-23.988	-26.584	-27.715	-20.370	-23.782	-27.229
Akaike Inf. Crit.	53.977	59.168	61.429	46.740	53.564	60.458
Note:						*p<0.1; **p<0.05; ***p<0.01

Table 25: Drought Logit regression Uruguay – 625 Km² resolution – Early period only

	Dependent variable:					
	Bad Years					
	(1)	(2)	(3)	(4)	(5)	(6)
LatAmEarly	-1.122**					
	(0.470)					
CHIRPSEarly		-0.165***				
		(0.059)				
CHIRPEarly			-0.181***			
			(0.066)			
TRMMEarly				-1.351***		
				(0.507)		
NDVIEarly					-32.354***	
					(11.400)	
EVIEarly						-33.654**
						(14.644)
Constant	1.398	2.880*	3.089*	2.130	17.988***	10.683**
	(1.232)	(1.495)	(1.605)	(1.321)	(6.903)	(5.364)
Observations	68	68	68	64	68	68
Log Likelihood	-21.604	-18.809	-19.260	-19.441	-21.737	-23.440
Akaike Inf. Crit.	47.208	41.618	42.521	42.883	47.475	50.880
Note:						*p<0.1; **p<0.05; ***p<0.01

Table 26: Excess Rainfall Logit regression Uruguay – 625 and 5,625 Km^2 resolution respectively

	Dependent variable:			
	Bad Years			
	(1)	(2)	(3)	(4)
LatAmEarly	0.206			
	(0.137)			
LatAmLate	0.313*			
	(0.176)			
CHIRPSEarly		0.016		
		(0.015)		
CHIRPSLate		0.045**		
		(0.022)		
CHIRPEarly			0.021	
			(0.021)	
CHIRPLate			0.031	
			(0.027)	
TRMMEarly				-0.027
				(0.145)
TRMMLate				0.365**
				(0.182)
Constant	-2.736***	-3.078***	-2.671***	-2.246**
	(0.828)	(0.951)	(1.017)	(0.885)
Observations	72	72	72	60
Log Likelihood	-37.810	-37.493	-39.107	-30.434
Akaike Inf. Crit.	81.620	80.986	84.214	66.868
Note:		*p<0.1; **p<0.05; ***p<0.01		

	Dependent variable:			
	Bad Years			
	(1)	(2)	(3)	(4)
LatAmEarly	0.211			
	(0.142)			
LatAmLate	0.323*			
	(0.183)			
CHIRPSEarly		0.016		
		(0.015)		
CHIRPSLate		0.045**		
		(0.022)		
CHIRPEarly			0.029	
			(0.021)	
CHIRPLate			0.030	
			(0.027)	
TRMMEarly				-0.033
				(0.149)
TRMMLate				0.367**
				(0.184)
Constant	-2.794***	-3.076***	-2.915***	-2.239**
	(0.855)	(0.948)	(1.040)	(0.895)
Observations	72	72	72	60
Log Likelihood	-37.823	-37.506	-38.658	-30.440
Akaike Inf. Crit.	81.647	81.012	83.315	66.880
Note:		*p<0.1; **p<0.05; ***p<0.01		

Table 27: Drought Logit regression Bolivia – 625 and 5,625 Km^2 resolution respectively

	Dependent variable:					
	Bad Years					
	(1)	(2)	(3)	(4)	(5)	(6)
LatAmEarly	-0.016					
	(0.216)					
LatAmLate	0.053					
	(0.259)					
CHIRPSEarly		-0.003				
		(0.023)				
CHIRPSLate		-0.053				
		(0.039)				
CHIRPEarly			-0.012			
			(0.029)			
CHIRPLate			0.001			
			(0.044)			
TRMMEarly				0.248		
				(0.339)		
TRMMLate				-0.450		
				(0.434)		
NDVIEarly					0.881	
					(8.231)	
NDVILate					-2.592	
					(3.954)	
EVIEarly						3.928
						(9.849)
EVILate						-0.112
						(5.642)
Constant	-1.338	1.293	-0.515	-1.172	-0.279	-3.172
	(1.417)	(1.332)	(1.316)	(2.714)	(6.114)	(6.367)
Observations	34	34	34	30	34	34
Log Likelihood	-18.528	-16.343	-18.338	-11.066	-18.332	-18.454
Akaike Inf. Crit.	43.056	38.687	42.675	28.132	42.664	42.908
Note:						*p<0.1; **p<0.05; ***p<0.01

	Dependent variable:					
	Bad Years					
	(1)	(2)	(3)	(4)	(5)	(6)
LatAmEarly	0.016					
	(0.247)					
LatAmLate	-0.049					
	(0.278)					
CHIRPSEarly		-0.018				
		(0.029)				
CHIRPSLate		-0.067				
		(0.044)				
CHIRPEarly			-0.021			
			(0.036)			
CHIRPLate			-0.001			
			(0.050)			
TRMMEarly				0.185		
				(0.360)		
TRMMLate				-0.471		
				(0.464)		
NDVIEarly					8.691	
					(10.090)	
NDVILate					-3.337	
					(4.725)	
EVIEarly						7.089
						(10.886)
EVILate						0.278
						(5.918)
Constant	-1.041	2.588	0.033	-0.627	-5.257	-4.947
	(1.587)	(1.723)	(1.633)	(3.054)	(6.679)	(6.380)
Observations	34	34	34	30	34	34
Log Likelihood	-18.534	-15.391	-18.138	-11.138	-17.977	-18.319
Akaike Inf. Crit.	43.068	36.781	42.277	28.277	41.955	42.638
Note:						*p<0.1; **p<0.05; ***p<0.01

Table 28: Drought Logit regression Chile – 625 and 5,625 Km^2 resolution respectively

	Dependent variable:					
	Bad Years					
	(1)	(2)	(3)	(4)	(5)	(6)
LatAmEarly	-1.237					
	(1.108)					
LatAmLate	-0.748					
	(0.834)					
CHIRPSEarly		-0.191				
		(0.140)				
CHIRPSLate		-0.161				
		(0.110)				
CHIRPEarly			-0.465**			
			(0.227)			
CHIRPLate			-0.191			
			(0.123)			
TRMMEarly				-1.111		
				(1.058)		
TRMMLate				-1.398		
				(1.131)		
NDVIEarly					-7.645	
					(9.407)	
NDVILate					48.668	
					(31.526)	
EVIEarly						-17.687
						(11.302)
EVILate						4.004
						(12.899)
Constant	1.066	4.786*	10.641**	2.144	-33.172	4.036
	(1.378)	(2.601)	(4.847)	(1.784)	(25.120)	(6.598)
Observations	34	34	34	30	32	32
Log Likelihood	-13.179	-11.637	-10.022	-8.153	-13.573	-13.919
Akaike Inf. Crit.	32.357	29.274	26.044	22.306	33.147	33.839
Note:						*p<0.1; **p<0.05; ***p<0.01

	Dependent variable:					
	Bad Years					
	(1)	(2)	(3)	(4)	(5)	(6)
LatAmEarly	-0.929					
	(1.036)					
LatAmLate	-0.898					
	(0.860)					
CHIRPSEarly		-0.154				
		(0.135)				
CHIRPSLate		-0.139				
		(0.101)				
CHIRPEarly			-0.188			
			(0.150)			
CHIRPLate			-0.116			
			(0.096)			
TRMMEarly				-1.084		
				(1.051)		
TRMMLate				-1.062		
				(0.941)		
NDVIEarly					-22.394	
					(13.856)	
NDVILate					32.288	
					(28.529)	
EVIEarly						-30.786*
						(18.136)
EVILate						17.772
						(19.232)
Constant	0.992	3.659	3.982*	1.592	-10.253	2.504
	(1.364)	(2.241)	(2.407)	(1.647)	(21.650)	(6.521)
Observations	34	34	34	30	32	32
Log Likelihood	-13.537	-11.891	-12.000	-8.927	-13.513	-13.348
Akaike Inf. Crit.	33.075	29.782	30.001	23.854	33.025	32.695
Note:						*p<0.1; **p<0.05; ***p<0.01

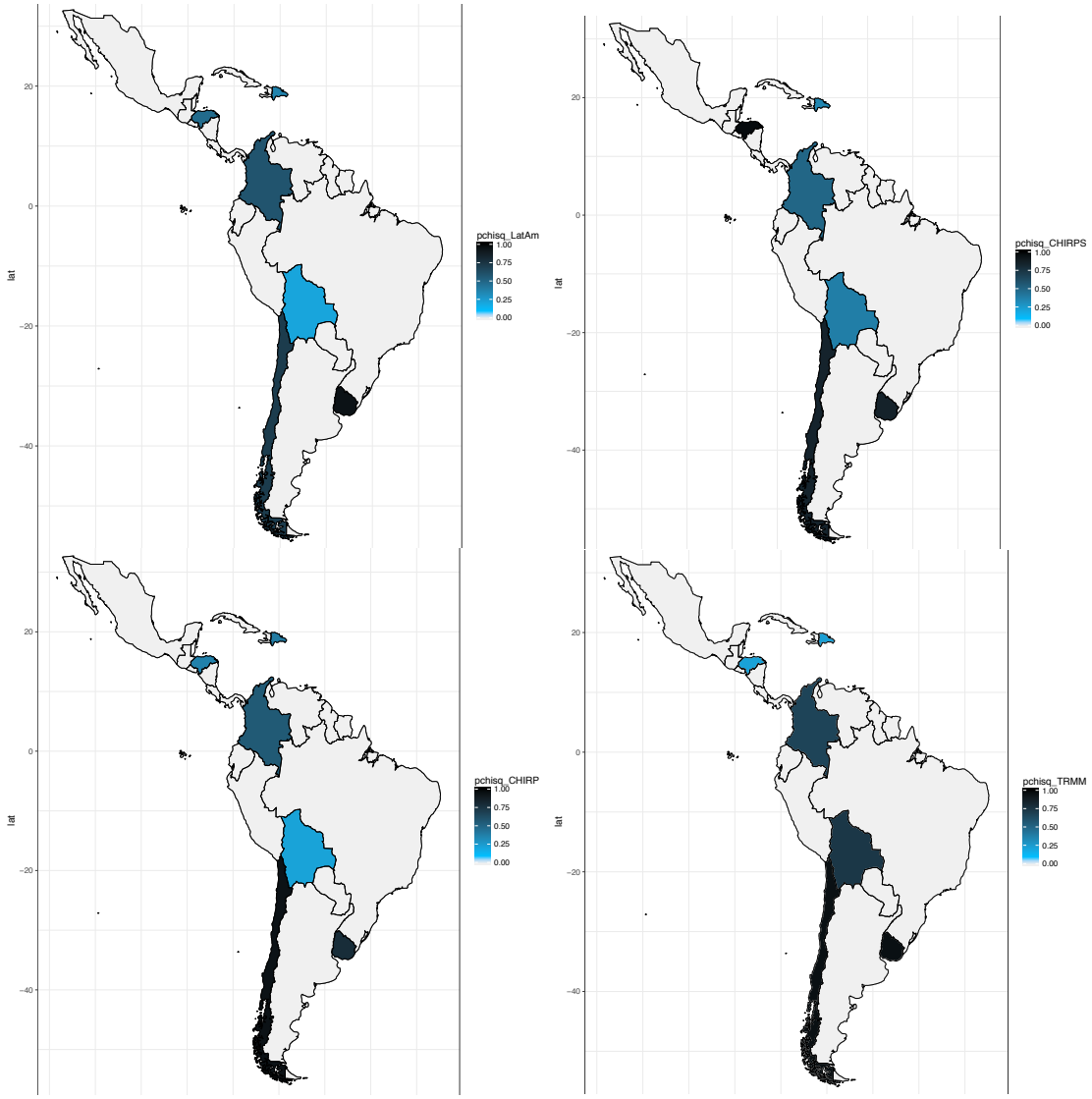
Table 29: Excess Rainfall Logit regression Chile – 625 and 5,625 Km^2 resolution respectively

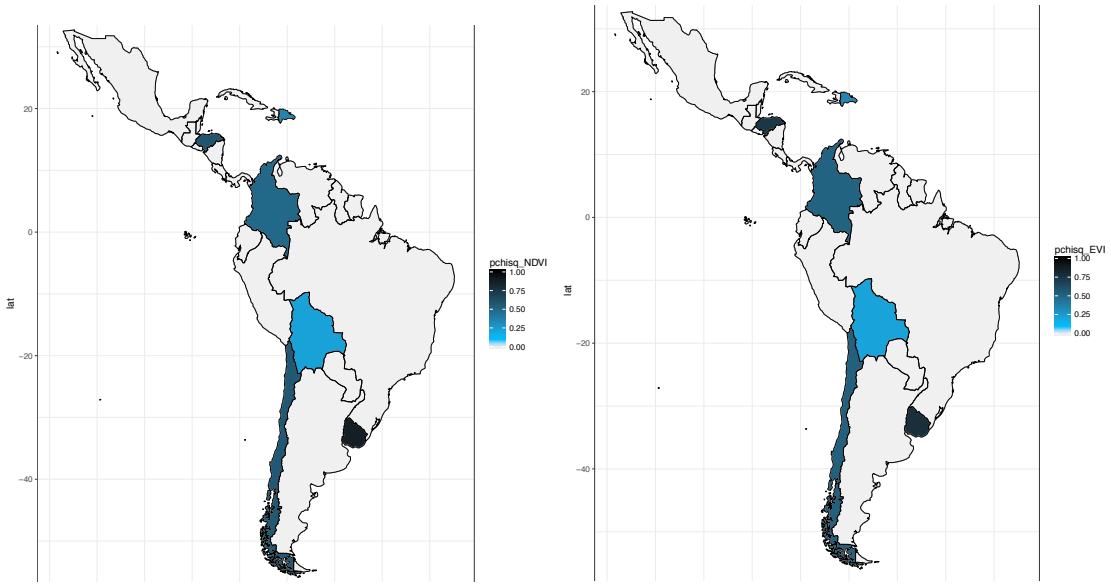
	Dependent variable:			
	Bad Years			
	(1)	(2)	(3)	(4)
LatAmEarly	-0.260			
	(0.332)			
LatAmLate	0.588			
	(0.419)			
CHIRPSEarly		-0.021		
		(0.036)		
CHIRPSLate		0.062		
		(0.040)		
CHIRPEarly			-0.008	
			(0.037)	
CHIRPLate			0.038	
			(0.028)	
TRMMEarly				-0.363
				(0.293)
TRMMLate				0.625
				(0.429)
Constant	-3.083*	-4.407	-3.650*	-3.343
	(1.855)	(2.693)	(2.079)	(2.421)
Observations	36	36	36	30
Log Likelihood	-16.577	-16.380	-16.716	-13.312
Akaike Inf. Crit.	39.153	38.760	39.432	32.624
Note:		*p<0.1; **p<0.05; ***p<0.01		

	Dependent variable:			
	Bad Years			
	(1)	(2)	(3)	(4)
LatAmEarly	-0.346			
	(0.401)			
LatAmLate	0.627			
	(0.502)			
CHIRPSEarly		-0.033		
		(0.038)		
CHIRPSLate		0.068		
		(0.043)		
CHIRPEarly			-0.051	
			(0.045)	
CHIRPLate			0.070*	
			(0.041)	
TRMMEarly				-0.701*
				(0.416)
TRMMLate				1.014*
				(0.594)
Constant	-2.709	-4.006*	-3.382*	-3.629
	(1.732)	(2.357)	(1.804)	(2.252)
Observations	36	36	36	30
Log Likelihood	-16.857	-16.211	-16.028	-12.668
Akaike Inf. Crit.	39.715	38.423	38.055	31.336
Note:		*p<0.1; **p<0.05; ***p<0.01		

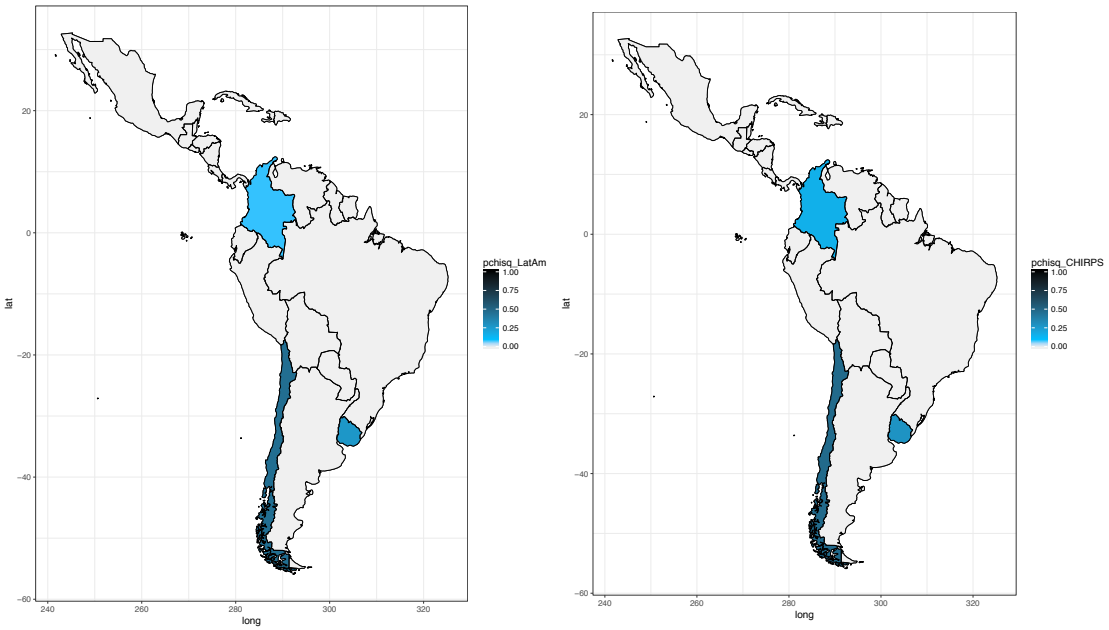
Annex 6
P-value Chi-square maps for each dataset

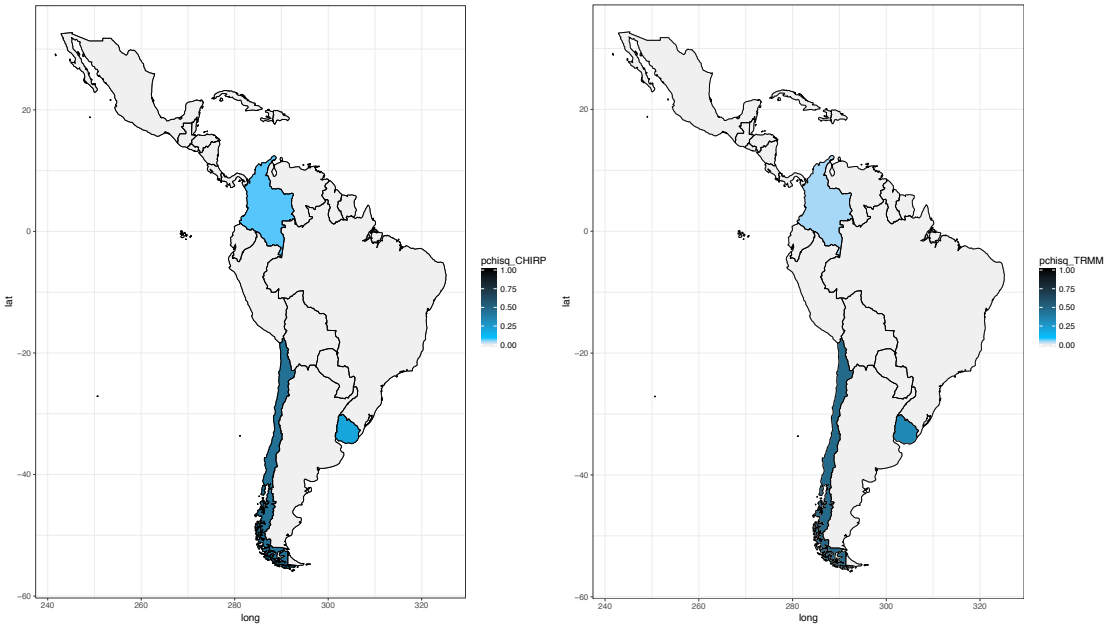
The p-value chi-square results for each dataset for the drought risk are displayed in the below maps. The first row contains the map corresponding to the LatAm and CHIRPS datasets, the middle row contains the map corresponding to the CHIRP and TRMM datasets and the bottom row contains the map corresponding to the NDVI and EVI datasets.





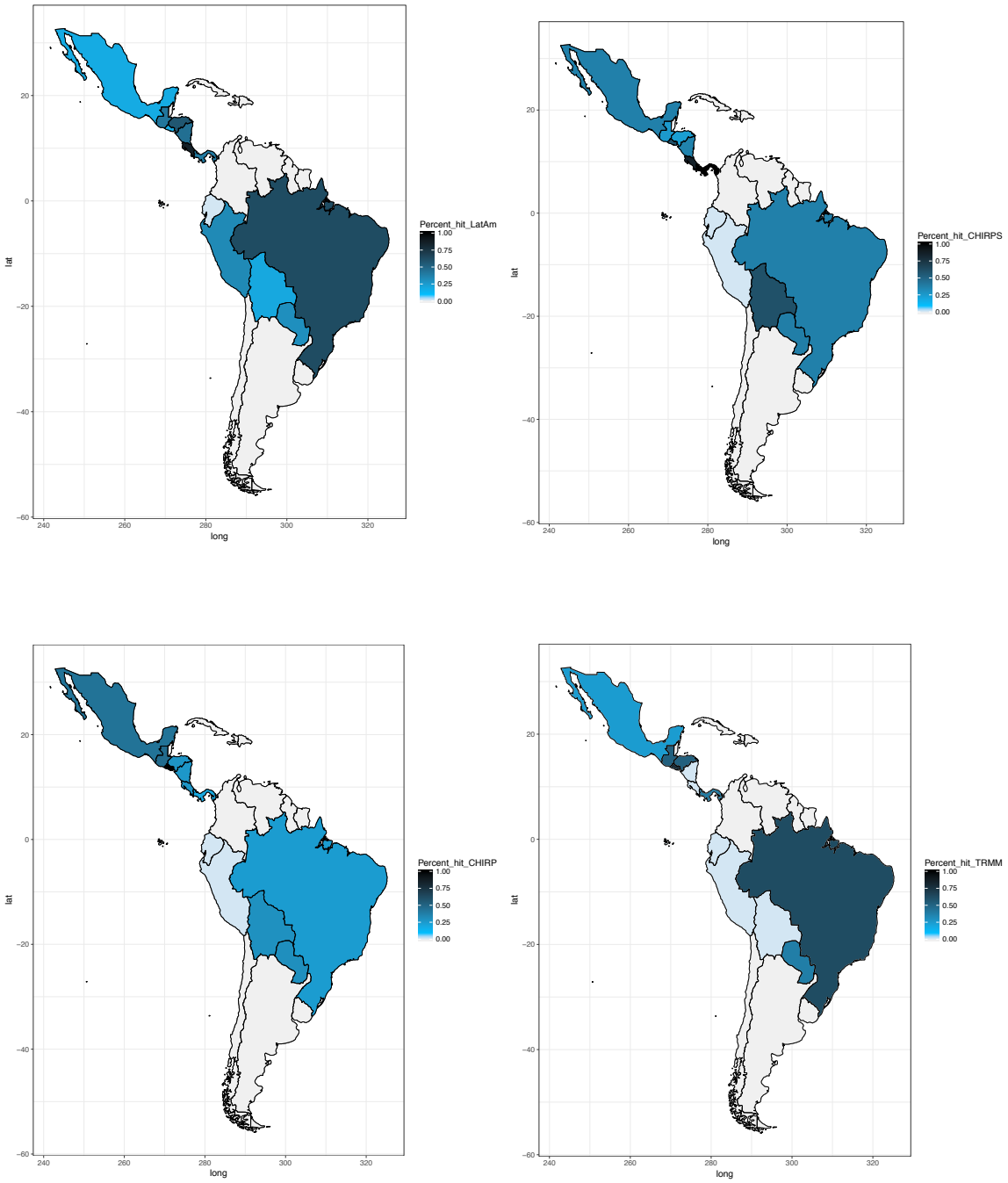
The p-value chi-square results for each dataset for the excess risk are displayed in the below maps. The first row contains the map corresponding to the LatAm and CHIRPS datasets and the bottom row contains the map corresponding to the CHIRP and TRMM datasets.




lat
long
pchisq_CHIRP
1.00
0.75
0.50
0.25
0.00
pchisq_TRMM
1.00
0.75
0.50
0.25
0.00

Annex 7
Percentage of hit maps for each dataset

The following maps show the percentage of hit of each dataset for drought events.



The following maps show the percentage of hit of each dataset for flood events.

